Synthetic Participants Generated by Large Language Models: A Systematic Literature Review

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Synthetic Participants Generated by Large Language Models: A Systematic Literature Review

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Abstract

In recent years, the prospect of Large Language Models (LLMs) for simulating participants within various research and data collection methods has been interrogated extensively. Its proponents cite aspirational promises, including high flexibility, adaptability, better representation and reduced research costs, all by leveraging the encoded wisdom of the internet crowd. Empirical studies paint a more nuanced but fragmented picture, with mixed results, heterogeneous methods and a saturation of different perspectives. In this systematic literature review, we delineate a clear and comprehensive conceptual understanding of LLM-generated participants and their comparative relationship to human samples. We synthesize the findings from 182 studies, obtained through a hybrid database and reference search, followed by a rigorous quality curation. Grounded in generalizable indicators, we present a standardized categorization of four fundamental issues that impact synthetic participants across diverse types of simulations – cognitive misalignments, distortions, misleading believability, and overfitting/contamination. Despite the survey revealing integrations of different LLMs, prompt engineering techniques, and participant or environment modeling methods, the fidelity improvements they demonstrated remain modest. At their most representative, LLMs may stochastically parrot data they were pre-trained on or fine-tuned with. To set appropriate expectations, explain their limitations and inform future applications, we propose the framing of synthetic participants as heuristic-like. Additionally, we discuss evaluation measures, specific supplemental roles that synthetic participants can be valid for, the underexplored potential of augmentative approaches, as well as a critical professional, social and ethical consideration of simulated insights.

Keywords: synthetic participants, large language models, participant simulation, prompt engineering, participant modeling, algorithmic fidelity

1 Introduction

Data from human participants is an integral source of truth across a multitude of fields, from social sciences to human-computer interaction. Businesses, organizations and governments all depend on behavioral or self-reported data from customer and user research, using platforms such as UXtweak[1] for insights to support their decision-making and policies. Vast human datasets are a precondition for many machine- and deep-learning solutions. Data collection, however, presents a variety of challenges that demand substantial effort to address (Rossi et al. 2024; Steinmacher et al. 2024). Recruiting audiences

[1]UXtweak user experience research platform: https://www.uxtweak.com//

with representative distributions and sufficient numbers can be difficult. Due to privacy, certain data types can only be collected in limited capacity, or not at all. Hardware requirements can make data collection infeasible in naturalistic environments (e.g., eye tracking (Kuric et al. 2025a)). Natural factors like fatigue may impede research and reduce the usefulness of its outputs. The accumulative costs of research processes may be substantial (Rossi et al. 2024; Steinmacher et al. 2024).

To tackle these issues, some scholars have suggested Large Language Models (LLMs) – generative artificial intelligence (AI) that can mimic humans by acting out elaborate and meaningful conversations – as surrogates for participants (Grossmann et al. 2023; Jansen et al. 2023; Argyle et al. 2023; Gerosa et al. 2024). If we could use synthetic participants to reliably approximate human thoughts and behavior, then vast amounts of new questions might be answered on-demand and at high volumes, making it easier to access large samples or zoom in on specific groups. The applicability of predictions in real-world settings could be improved by simulating complex and variable conditions (e.g., different devices, distracting surroundings) (Kuric et al. 2025c,b). The privacy, scalability, flexibility, adaptability and cost-efficiency of research could be revolutionized (Tzachristas et al. 2025). The key caveat lies in the extent to which algorithmically generated data can actually reflect real data collection methods and populations. This is referred to as algorithmic fidelity (also alignment, such as behavioral or psychometric, in some works) (Amirova et al. 2024; Argyle et al. 2023; Mori et al. 2024; Xie et al. 2024; He-Yueya et al. 2024).

Some scholars accurately acknowledge that LLMs differ from humans by lacking the same psychological and neurophysiological processes, as well as personal beliefs, goals or desires (Hämäläinen et al. 2023; Shanahan et al. 2023). From the behavioral standpoint, LLMs were examined in various settings by assuming the role of participants (Xie et al. 2024; Wang et al. 2025d). The current gap lies in a lack of multidisciplinary research that would analyze the full scale of the current literature, its patterns, trends and conclusions. For example, failing to replicate psychological experiments on reasoning may have deep downstream implications for interpreting results in other fields. We aim to provide a comprehensive and generalizable conceptual framework that clarifies the potentials and limitations of synthetic participants. To examine methodological avenues (e.g., claims that the missing cognitive processes, desires, etc. can be simply prompt-engineered into the LLMs) and identify promising ones, we will also explore the various implemented techniques and assess their effectiveness. As we show in Table 1, the most closely related works either do not represent rigorous systematic literature reviews, are research agendas, perspective articles, their scope is limited, or their thematic relevance is tangential (e.g., persona generation).

Of further significance is that this review presents a timely response to the current high valuation of technology firms centered on AI[2], contrasted with low AI revenues and limited consumer adoption (Challapally et al. 2025). Amidst the current social climate, where businesses are pressured by stakeholders to leverage AI to reduce their operating costs (including research costs), the practical importance of deep insights about synthetic participants, independent of tech corporations' interests, rises significantly.

Our article presents the following contributions:

- A systematic literature review, integrating the findings from 182 high-quality publications involving LLM-generated synthetic participant data.
- Detailed summary of settings (domains and tasks) where synthetic participants have been explored.
- Rigorous multidisciplinary thematic synthesis of issues associated with synthetic participants, categorized into four perspectives:
  - Misalignment with cognitive processes,
  - Distortions and biases,
  - Misleading believability,
  - Overfitting and contamination.
- Analysis of the effectiveness of techniques for improving synthetic participants (prompt engineering, participant modeling, model selection).
- Refined concept of synthetic participants as a supplemental heuristic-like approach, featuring theoretical explanations and methodological suggestions for practitioners.

[2]fortune.com – Without data centers, GDP growth was 0.1% in the first half of 2025, Harvard economist says: https://fortune.com/2025/10/07/data-centers-gdp-growth-zero-first-half-2025-jason-furman-harvard-economist/

Table 1 Prior secondary research relevant to synthetic research participants. Asterisks (*) denote preprints.

Reference	Year	Focus	SLR	Corpus	Limitations
Ours	2025	LLM-generated synthetic participants	Yes	182	-
Amin et al. (2025)	2025*	AI in persona development	Yes	52	Marginal relevance, focus on persona generation
Sufi (2024)	2024	Augmentation and generation of research data	Yes	77	Limited coverage of synthesis in favor of enhancement, including data other than simulation of human subjects
Goyal and Mahmoud (2024)	2024	Generation of synthetic data	Yes	77	Focus on diverse types of data (images, tabular, speech, textual) rather than synthetic participants and their human likeness
Peng and Yang (2025)	2025*	LLMs as tools in social research	No	-	Social research only, focus on general use of LLMs, not a systematic literature review
Anthis et al. (2025)	2025*	LLMs as agents in social simulations	No	-	Position paper, does not analyze current findings from studies implementing directions suggested as promising
Gerosa et al. (2024)	2024	Vision for LLM-driven synthetic participants in software engineering, discussion of open problems	No	-	Vision article of limited scope, does not provide systematic review of studies and findings
Rossi et al. (2024)	2024	Perspectives of and issues with LLM-generated data in social science research	No	-	Essay rather than systematic literature review, scope is only social science
Sarstedt et al. (2024)	2024	Application of synthetic silicon samples in marketing	No	-	Focus on customer research, does not systematically review studies and findings
Demszky et al. (2023)	2023	LLMs as tools in psychology	No	-	Marginal relevance (generated responses to stimuli), focus on general LLM capabilities and limitations

Content summary. Section 2 defines our research questions. Section 3 details the methodology by which our systematic literature review was conducted. Section 4 presents the results of our analysis. Section 5 discusses the key implications of our findings for the concept and applications of synthetic participants. Section 6 transparently declares the threats to the validity of our findings. Section 7 presents an overview of our key conclusions.

Definitions. For conciseness and considering our subject, when we discuss *synthetic participants*, we are referring to LLM-generated synthetic participants, unless explicitly stated otherwise. The less common synonym *silicon participants*, primarily found in social science publications, may also be used.

2 Research questions

Our review questions the degree to which LLMs can simulate participants according to the current state of evidence. Largely in part to the territory's technological immaturity and its thematic and granular heterogeneity, we discern six research questions as the foundations for organizing nuanced contextual, empirical and methodological insights (see Table 2). Our research questions can be divided into three blocks:

- Context of application. What domains (RQ1) and tasks (RQ2) were synthetic participants applied to (e.g., healthcare, surveys).
- Limiting factors and properties. What observed issues (RQ3) empirically hindered their effectiveness (e.g., different types of algorithmic biases).
- Methods and technologies. What methods and technologies, including prompt engineering (RQ4), participant modeling (RQ5), and model selection (RQ6), were used by the intended solutions (e.g., chain-of-thought, personas incorporating age and gender, GPT-4).

Table 2 Research questions investigated by our systematic literature review.

#	Title	Question
RQ1	Domain	In what domains are synthetic participants studied?
RQ2	Task	What types of task scenarios are synthetic participants applied to?
RQ3	Issues	What issues are associated with synthetic participants?
RQ4	Prompt engineering	What prompting strategies are used to generate synthetic participants?
RQ5	Participant modeling	What participant modeling strategies are used in LLM generation of synthetic participant data?
RQ6	Model selection	What LLMs are used to generate synthetic participants?

3 Methodology

We followed the guidelines outlined by Carrera-Rivera et al. (2022) for systematic literature reviews in computer science. Based on these guidelines, we collected publications, removed duplicates and conducted an initial screening by reviewing titles and abstracts. At this stage, reference search was conducted to mitigate selection bias. Finally, the selected publications were examined in detail, and relevant data was extracted. See the flow diagram in Figure 1 for the protocol and the number of publications included in its steps.

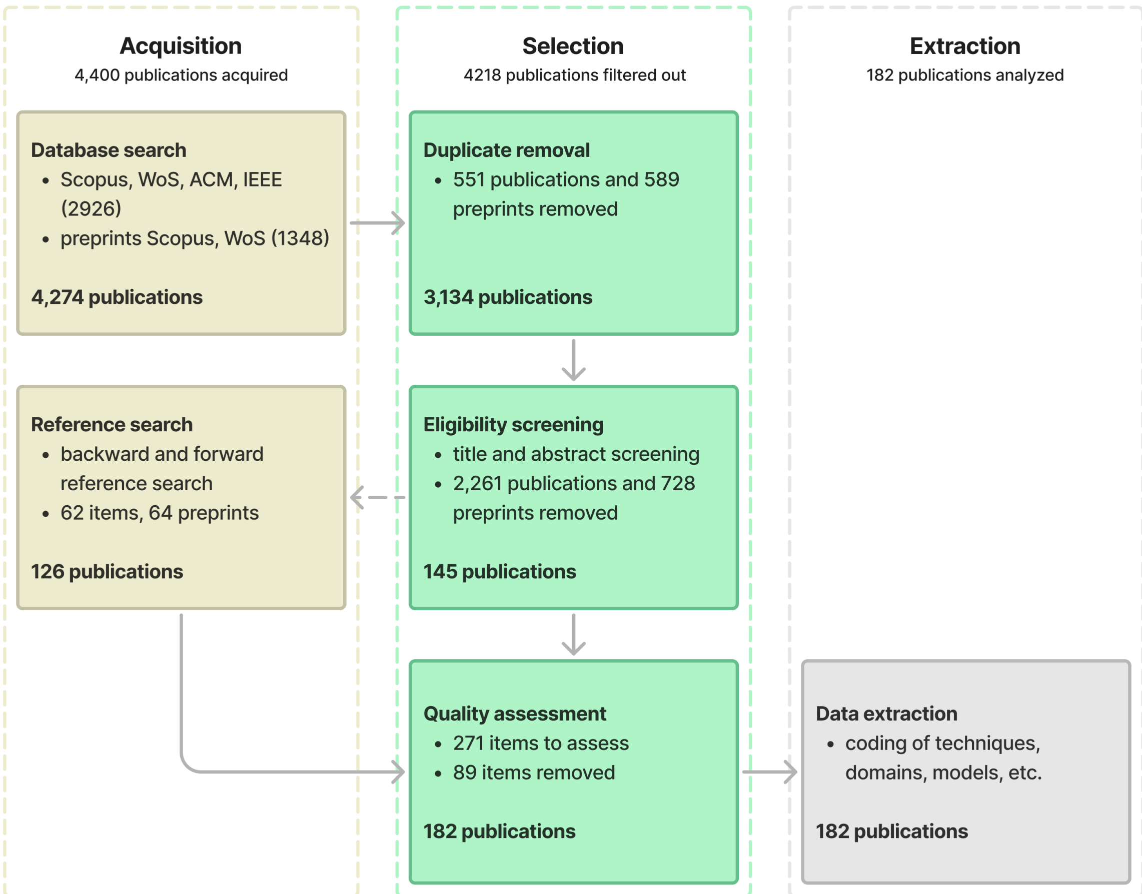

Fig. 1 Flow diagram of the systematic literature review protocol representing a hybrid of keyword and reference search.

To mitigate researcher bias, all activities linked to the review of publications were performed by two researchers. Disagreements were resolved through discussion. A third researcher was assigned the role of adjudicator, although in practice, consensus was always reached before their input was needed.

3.1 Database search

Among academic literature databases, we selected those with the most extensive multidisciplinary coverage (Scopus[3], Web of Science[4]), alongside discipline-specific databases in engineering and computer science (IEEE Xplore Digital Library[5], ACM Digital Library[6]). These databases offer advanced search capabilities representative of their status as standard sources of information in related literature reviews (Wei et al. 2024; Sufi 2024; Carrera-Rivera et al. 2022). Our queries were executed on 30/06/2025. The inclusion criteria focused on publications in the English language, categorized as journal articles, conference papers or preprints. Preprints were included in the analysis to maximize the timeliness of our review.

Keywords were tuned iteratively on 15 relevant articles acquired manually. They included wildcards to match diverse phrasing used by authors. Final keywords (see Table 3 for library-specific implementations) captured the central concept of our research through three defining aspects:

- Data of artificial origins: synthetic, simulated, generated
- Representation of humans: participant, respondent, response, user, human, research, person, data
- Use of LLMs: LLM, large language model, GPT, generative AI

Full bibliographic entries were extracted for published papers and articles, while only abstracts, titles and author names were available for preprints.

Table 3 Implementation of database queries within databases and the number of yielded publications.

Database	Query	Publications	Preprints
Scopus	TITLE-ABS-KEY ((”synthetic” OR ”simulat*” OR ”generate?” OR ”AI-generate?” OR ”LLM-generate?”) W/3 (”participant?” OR ”respondent?” OR ”response?” OR ”user?” OR ”human?” OR ”research” OR ”person*” OR ”data” OR ”data generation”) W/3 (”large language model” OR LLM OR GPT OR ChatGPT OR ”language model” OR ”generative AI”))	1188	993
WoS	TS=(((”synthetic” OR ”simulat*” OR ”generate?” OR ”AI-generate?” OR ”LLM-generate?”) NEAR/3 (”participant?” OR ”respondent?” OR ”response?” OR ”user?” OR ”human?” OR ”research” OR ”person*” OR ”data” OR ”data generation”) NEAR/3 (”large language model” OR LLM OR GPT OR ChatGPT OR ”language model” OR ”generative AI”)))	517	355
IEEE	((”All Metadata”:”synthetic” OR ”All Metadata”:”simulat*” OR ”All Metadata”:”generate?” OR ”All Metadata”:”AI-generate?” OR ”All Metadata”:”LLM-generate?”) NEAR/3 (”All Metadata”:”participant?” OR ”All Metadata”:”respondent?” OR ”All Metadata”:”response?” OR ”All Metadata”:”user?” OR ”All Metadata”:”human?” OR ”All Metadata”:”research” OR ”All Metadata”:”person*” OR ”All Metadata”:”data” OR ”All Metadata”:”data generation”) NEAR/3 (”All Metadata”:”large language model” OR ”All Metadata”:LLM OR ”All Metadata”:GPT OR ”All Metadata”:ChatGPT OR ””All Metadata”:language model” OR ”All Metadata”:”generative AI”))	245	-
ACM	Abstract:((”synthetic” OR ”simulat*” OR ”generate?” OR ”AI-generate?” OR ”LLM-generate?”) AND (”participant?” OR ”respondent?” OR ”response?” OR ”user?” OR ”human?” OR ”research” OR ”person*” OR ”data” OR ”data generation”) AND (”large language model” OR LLM OR GPT OR ChatGPT OR ”language model” OR ”generative AI”)) OR Title:((”synthetic” OR ”simulat*” OR ”generate?” OR ”AI-generate?” OR ”LLM-generate?”) AND (”participant?” OR ”respondent?” OR ”response?” OR ”user?” OR ”human?” OR ”research” OR ”person*” OR ”data” OR ”data generation”) AND (”large language model” OR LLM OR GPT OR ChatGPT OR ”language model” OR ”generative AI”)) OR Keyword:((”synthetic” OR ”simulat*” OR ”generate?” OR ”AI-generate?” OR ”LLM-generate?”) AND (”participant?” OR ”respondent?” OR ”response?” OR ”user?” OR ”human?” OR ”research” OR ”person*” OR ”data” OR ”data generation”) AND (”large language model” OR LLM OR GPT OR ChatGPT OR ”language model” OR ”generative AI”))	976	-
Sum		2926	1348

[3]Scopus: https://www.scopus.com/
[4]Web of Science: https://www.webofscience.com/
[5]IEEE XPlore Digital Library: http://ieeexplore.ieee.org/
[6]ACM Digital Library: https://dl.acm.org/

3.2 Duplicate removal

The `pybtex` library was used to import `.bib` files containing bibliographic data of 4,274 publications. Faulty entries, such as keys containing spaces, were corrected, and data formats were standardized. The files were then merged, with duplicates identified and removed by matching DOIs, author names or similar titles. Additionally, since preprints and their published versions may contain differences, similarity scores for their titles and author lists were calculated using the fuzzy string matching library `rapidfuzz`. These were used to manually compare publications with higher similarity. After deduplication, 3,134 unique publications remained.

3.3 Eligibility screening

Keyword search results contained a high ratio of irrelevant publications – e.g., healthcare studies where real people role-played as patients, expert perspectives about the ability of LLMs to respond to questions expertly, or the generation of different types of synthetic datasets for neural networks. To obtain highly relevant publications, a two-stage LLM-assisted eligibility screening procedure was implemented.

During the first stage, publications were prescreened with GPT-4. An iteratively tuned prompt (shown in Appendix A) was verified to reject obviously irrelevant publications and remain lenient in the case of ambiguity. Forty publications were labeled per batch. To mitigate order bias, batches were selected randomly. Filters were executed three times per batch to increase specificity, using outputs from the previous iteration as inputs. The second stage was manual. The titles and abstracts of the 20% accepted by LLMs were reviewed carefully. Among the 80% of articles rejected by the LLM, a title-based validation revealed zero false negatives. This screening reduced the size of the corpus to 145 publications.

3.4 Reference search

As a safeguard against missing relevant research due to keyword bias, we performed backward and forward reference search in 145 publications. This approach is derived from single-iteration snowball search (Wohlin et al. 2022; Mourão et al. 2020). We collected and exported the literature with Zotero[7], then reviewed the titles and abstracts. As the result of the reference search, 126 additional publications were obtained (36 articles, 26 papers, 64 preprints), resulting in a total sample of 271 publications.

3.5 Quality assessment

The contents of 271 articles were reviewed to establish high quality and methodological rigor of analyzed research and the reliability of its findings. See Table 4 for definitions of the inclusion criteria. Three criteria (1-3) were applied universally, while a fourth criterion was applied based on publication status. The credibility of published papers and articles was assessed based on the journal or conference ranking (4a). For preprints, a quality checklist was applied instead, given their lack of a peer review (4b). Depending on whether the criteria were met, scores were assigned as a binary 1 or 0. A full score of 4 was required for inclusion in the final corpus. The assessment yielded the final set of 182 publications.

3.6 Data extraction

Data for answering the research questions was coded concurrently with data used for the quality assessment. Given the open-ended nature of the questions, coding for domains, tasks or issues was adjusted iteratively as recurring patterns were identified with deepening understanding of the literature. For participant modeling, a coding based on the taxonomy of single-persona (renamed from persona-based since all approaches may be classified based on their representation of personas, even if implicitly), multi-persona, and mega-persona introduced by Gerosa et al. (2024). Notably, personas are differentiated by the sample unit that they represent in a simulation (an individual, group or population), even if said unit does not specify its attributes that would be associated with a traditional persona.

[7]Zotero research assistant tool: https://www.zotero.org/

Table 4 Quality assessment criteria applied in the final manual review of the article as a whole.

#	Criterion	Description
1	Relevance	Relevance to the research questions – whether the research concerned the use of synthetic participants generated through LLMs.
2	Experiment	Whether an experiment was conducted, involving LLM synthetic participants. This excludes publications,secondary literature and theoretical papers.
3	Availability	Availability of full research document online, adhering to the principles of Open Science: transparency and sharing.
4a	Source credibility (published articles / papers only)	The quality of the journal or conference was considered. Journal Citation Report[1] rank of Q3 and higher (Q1–Q3), or ICORE[2] conference ranks of B and higher (A*, A, B) were qualified. As fallback for missing entries, Scimago[3] Journal Rank and Conference ranks[4] were used.
4b	Quality checklist (preprints only)	A checklist was developed based on existing quality ranking checklists (Carneiro et al. 2020; Bossuyt et al. 2015; von Elm et al. 2007; Hair et al. 2019; Kilkenny et al. 2010; Moher et al. 2001). Preprints scoring at least 15 out of the 19 items were assigned a quality score of 1. This threshold was determined upon manual review of the preprints' contents. The checklist includes research goal information, experimental design and clarity, sample details, statistical reporting, interpretation of findings, and disclosures on research transparency (see Appendix B).

[1]Journal citation report: https://jcr.clarivate.com/

[2]ICORE Conference portal: https://portal.core.edu.au/conf-ranks

[3]Scimago journal rank: https://www.scimagojr.com/

[4]Conference ranks: http://www.conferenceranks.com/

4 Results

We organize our findings based on analyses of the posed research questions. RQ1 and RQ2 illustrate the context under which LLM-synthesized participants were studied by analyzing distributions of tasks and domains. RQ3 through RQ6 provide detailed analysis of incorporated approaches and their effectiveness, organized thematically to identify occurrence of patterns across different fields.

4.1 Domain (RQ1)

RQ1: In what domains are synthetic participants studied?

The possibility of synthesizing participant data was the subject of exploration in a diversity of fields, as shown by the distribution of domains in Figure 2. Predictably, the most prevalent fields are strongly human-centric: social & political sciences, psychology and human-computer interaction. The use cases of synthetic participants differ thematically between domains, as summarized in Table 5. In this section, we analyze the domain distribution. For a detailed analysis of incorporated approaches and their effectiveness, organized thematically to identify patterns across different fields, see 4.3 Issues through 4.6 Model selection.

4.1.1 Social & Political Science

The most frequent use of synthetic participants in socio-political fields involves simulation of public opinion. Prediction of voting in elections was studied in national contexts of the USA (Argyle et al. 2023; Bisbee et al. 2024; Zhang et al. 2024c) and Germany (Heyde et al. 2025), as well as multinationally (Qi et al. 2025; Qu and Wang 2024; Shrestha et al. 2024). Other studies investigated the ability of LLMs to replicate surveys of human values and preferences (social, economic, political, moral, religious) (Qu and Wang 2024; Durmus et al. 2024; Santurkar et al. 2023; Kambhatla et al. 2025; Park et al. 2024b; Giorgi et al. 2024), perceived value of occupations (Gmyrek et al. 2025), perceived offensiveness and toxicity (Sun et al. 2024, 2025a; Giorgi et al. 2024) and demographics (Dominguez-Olmedo et al. 2024). Instruments such as the Political Compass Test were used to assess model biases (Hartmann et al. 2023; Rozado 2023; Rutinowski et al. 2024; Bernardelle et al. 2025; Boelaert et al. 2025).

Surveys were studied in the general social context (Wang et al. 2025a; Jiang et al. 2025; Park et al. 2024a), as well as with specific focus on labor policy (Lee et al. 2024b) organizational settings (Suguri Motoki et al. 2024) and misinformation (Neumann et al. 2025). Using historical social survey

Table 5 Synthetic participant generation literature categorized by domain.

Domain	Synthetic participant use cases	Publications
Social & Political Science	Prediction of elections, surveys of public opinion, beliefs and social values, social interaction simulation.	Argyle et al. (2023); Ball et al. (2025); Bernardelle et al. (2025); Bisbee et al. (2024); Boelaert et al. (2025); Busker et al. (2025); Cheng et al. (2023); Cho et al. (2024); Chuang et al. (2024); Dominguez-Olmedo et al. (2024); Durmus et al. (2024); Flamino et al. (2025); Giorgi et al. (2024); Gmyrek et al. (2025); Hartmann et al. (2023); Heyde et al. (2025); Huang et al. (2025a); Hwang et al. (2023); Jiang et al. (2025); Kalinin (2023); Kambhatla et al. (2025); Kang et al. (2025); Kim and Lee (2024); Lee et al. (2024a,b, 2025); Li et al. (2025); Ma et al. (2025a); Manning et al. (2024); Matsumura et al. (2025); Motoki et al. (2024); Neumann et al. (2025); Park et al. (2024a,b); Qi et al. (2025); Qu and Wang (2024); Rozado (2023); Rutinowski et al. (2024); Sanders et al. (2023); Santurkar et al. (2023); Shrestha et al. (2024); Suguri Motoki et al. (2024); Suh et al. (2025); Sun et al. (2024, 2025a); Wang et al. (2025a); Zhang et al. (2024c,d); Zhou et al. (2025)
Psychology	Simulation of self-reported and behavioral results linked to personality, emotion, reasoning, cognitive skills and biases.	Aher et al. (2023); Almeida et al. (2024); Amirizaniani et al. (2024); Bhandari et al. (2025); Binz and Schulz (2023); Bodroza et al. (2024); Cava and Tagarelli (2025); Dey et al. (2025); Dorner et al. (2023); Elyoseph et al. (2023); Griffin et al. (2023); Hagendorff et al. (2023); Han et al. (2024); Havaei et al. (2024); Heyman and Heyman (2024); Huang et al. (2024a,b,c,d); Huff and Ulakçı (2024); Jiang et al. (2023, 2024); Jiaqi et al. (2025); Kovač et al. (2024); Kwok et al. (2024); Li and Qi (2025); Lim et al. (2025); Lo et al. (2025); Ma et al. (2024); Marjieh et al. (2024); Milička et al. (2024); Molchanova et al. (2025); Niszczota et al. (2025); Park and Kim (2024); Pellert et al. (2024); Petrov et al. (2024); Salecha et al. (2024); Serapio-García et al. (2025); Sparrenberg et al. (2024); Tavast et al. (2022); Tjuatja et al. (2024); Trott (2024); Wang et al. (2025d,e); Winter et al. (2024); Zhang et al. (2025)
Human-Computer Interaction & User Experience	Simulated user research and user-centered design methods, persona generation, data for improving advanced user interfaces.	Al et al. (2024); Algherairy and Ahmed (2025); Ataei et al. (2025); Barambones et al. (2024); Benharrak et al. (2024); Fang et al. (2024); Gu et al. (2025); Gupta et al. (2025); Hämäläinen et al. (2022, 2023); Hu et al. (2023); Hung et al. (2025); Jung et al. (2025); Kaate et al. (2025); Kapania et al. (2025); Khanshan et al. (2024); Lazik et al. (2025); Li et al. (2024a,c); Lu et al. (2025); Ma et al. (2025b); Nóbrega et al. (2025); Park et al. (2022, 2023); Salminen et al. (2024); Sattele and Ortiz (2024); Schuller et al. (2024); Sethi et al. (2025); Shin et al. (2024); Sun et al. (2025b); Venkit et al. (2025); Xiang et al. (2024); Yang et al. (2025); Zhang et al. (2024b)
Computer Science & Engineering	Information retrieval, recommendation systems, Internet of Things, computational linguistics, software engineering surveys.	Alaofi et al. (2023); Bakaev et al. (2025); Braga et al. (2024); Du et al. (2024); Fiori et al. (2024); Huang et al. (2025b); Leng et al. (2025); Owoicho et al. (2023); Ran et al. (2025); Robinson et al. (2023); Sannigrahi et al. (2024); Steinmacher et al. (2024); Wang et al. (2025b); Xu et al. (2025); Yonekura et al. (2024); Yoon et al. (2024); Zhang et al. (2024a); Zhao et al. (2025); Zhu et al. (2025)
Education	Exams, essays, behavior while learning coding, training conversations.	Abbasiantaeb et al. (2024); Benedetto et al. (2024); He-Yueya et al. (2024); Herbold et al. (2023); Khalil et al. (2025); Liu et al. (2025); MacNeil et al. (2024); Min and Jeong (2024); Zhan et al. (2025)
Healthcare	Patient counseling, interviews and interventions, e-health records and testing data.	Abdel-Khalek et al. (2025); Amirova et al. (2024); Anderson et al. (2024); De et al. (2025); Gray et al. (2024); Jalali and Akhavan (2024); Mori et al. (2024); Smrke et al. (2025); Wang et al. (2024)
Mobility & Tourism	Mobility tracking in metropolitan areas, travel and tourism surveys.	Bhandari et al. (2024); Homer (2025); Ivasciuc et al. (2025); Ju et al. (2025); Sop and Kurcer (2024); Tzachristas et al. (2025)
Marketing	Consumer preferences, perception of brands and product attributes, willingness to pay.	Bickley et al. (2025); Brand et al. (2024); Imschloss et al. (2025); Li et al. (2024b); Shan and Jia (2024); Wang et al. (2025c)
Economics	Economic behavior, strategies, knowledge and experiences.	Alonso et al. (2025); Gao et al. (2025); Kitadai et al. (2024); Xie et al. (2024)

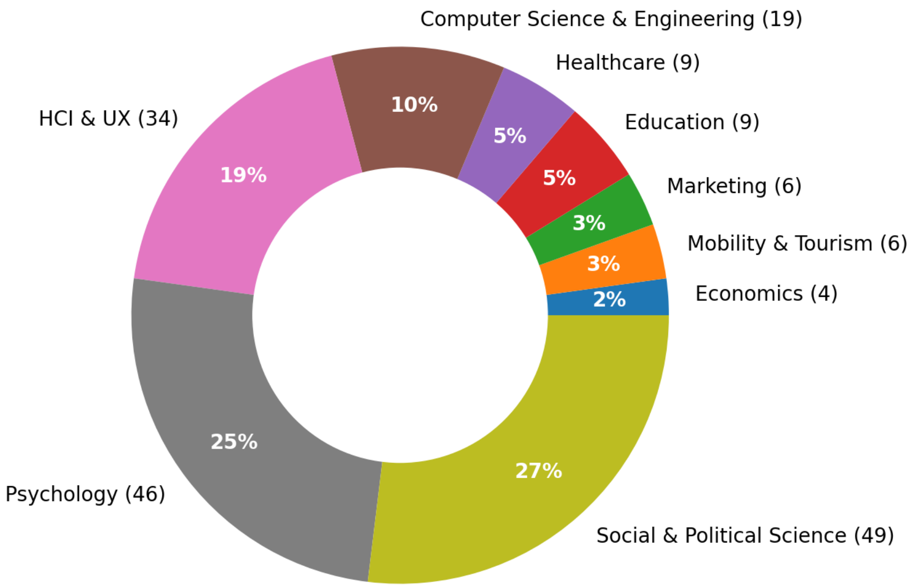

Fig. 2 Distribution of domains where LLM-driven synthetic participants were studied.

datasets, Kim and Lee (2024) developed a framework to predict individuals' missing responses and responses to unasked questions based on a model of their beliefs and temporal context. LLM-generated personas were applied in socio-political data simulations (Suguri Motoki et al. 2024; Bernardelle et al. 2025; Kalinin 2023). Properties of personas created by LLMs were investigated by Cheng et al. (2023), whereas Busker et al. (2025) assessed LLM-generated stereotypes. Huang et al. (2025a) presented an approach for adaptive sample size determination based on quantified uncertainty and statistical misalignment with the population.

Social interaction ability of LLMs was also assessed in discussion scenarios, including conflict resolution (Matsumura et al. 2025), auctions, bail hearings and job interviews (Manning et al. 2024), as well as debate-based games (Flamino et al. 2025).

4.1.2 Psychology

Through models of the human psyche and insights about mental processes, psychology brings broad cross-domain implications for synthetic participants. Analogous probing of the minds of human participants and LLMs is invaluable for comparative and bidirectional analysis. Despite the mechanistic disparities from humans and their psychological processes (Pellert et al. 2024), terms such as cognition, views and emotions are used as convenient metaphors in machine psychology for studying LLM behavior. By channeling psychological theory that explains human behavior and self-reporting, better alignment between synthetic and human output could be achieved. A substantial portion of psychological characteristics that affect behavior are subjective, with no wrong or correct answers. These include personal values (Kovač et al. 2024; Lim et al. 2025; Pellert et al. 2024), interpersonal relationships (Huang et al. 2024c), motivations (Zhang et al. 2025; Huang et al. 2024c) and self-construal (Li and Qi 2025).

Personality traits, which capture differences in attitude between people, were unequivocally the most abundantly studied psychological factor in participant synthesis. Occurrence and manifestation of personality in LLM output was examined alongside the ability of LLMs to simulate personality types, which achieved mixed results (Bhandari et al. 2025; Cava and Tagarelli 2025; Dey et al. 2025; Dorner et al. 2023; Han et al. 2024; Huang et al. 2024a,c,d; Jiang et al. 2023, 2024; Lim et al. 2025; Li and Qi 2025; Lo et al. 2025; Molchanova et al. 2025; Niszczota et al. 2025; Pellert et al. 2024; Petrov et al. 2024; Salecha et al. 2024; Serapio-García et al. 2025; Sparrenberg et al. 2024; Wang et al. 2025d,e; Winter et al. 2024). Aside from Big Five as the most commonly used personality model, some research used MBTI (Cava and Tagarelli 2025; Jiaqi et al. 2025), SLOAN (Havaei et al. 2024), as well as personality-adjacent constructs such as affect, aggression and creative self-concept (Petrov et al. 2024).

Cognitive traits and abilities focused on by research include cognitive reflection, decision making, deliberation, causal reasoning, moral and legal reasoning and memory (Hagendorff et al. 2023; Binz and Schulz 2023; Almeida et al. 2024; Huff and Ulakçı 2024). Emotional awareness and responses represent a linked area of interest (Huang et al. 2024b; Park and Kim 2024; Tavast et al. 2022; Elyoseph et al. 2023) for understanding machine behavior. Theory of Mind – the ability to model mental states through factors including intentions, percepts, beliefs and emotions as foundations of reasoning – was studied as a comprehensive theoretical framework for assessing the higher reasoning capabilities of LLMs (Amirizaniani et al. 2024). That includes simulation of its development in children (Milička et al. 2024).

Other studies focused on replication human response biases (Tjuatja et al. 2024) and of empirically observed psychological effects with LLMs. Aher et al. (2023) proposed a framework titled Turing Experiment for assessing the realism of human simulation, demonstrated on four classical behavioral experiments (e.g., Milgram Shock Experiment). LLMs were assessed in psycholinguistic tasks heyman2024, with Trott (2024) measuring the number of humans needed to outperform them by sample quality. Social psychology assessed the effects of social influence, persuasion and misinformation in consumed media or populistic rhetoric based on cultural and demographic attributes (Kwok et al. 2024; Ma et al. 2024), as well as past exposure to false information (Griffin et al. 2023).

4.1.3 Human-Computer Interaction & User Experience

At the intersection of human factors, technology and design, feedback from human participants represents the golden standard for understanding and empathizing with stakeholders. The pressure toward fast design and development of increasingly complex and adaptive user experiences drives the demand for rapidly-available and reliable sources of data.

A part of synthetic participant research mirrors the major paradigms of user research. Behavior was simulated in usability testing to detect usability issues that harm user experience, or in Wizard of Oz studies to investigate behavioral expectations (Xiang et al. 2024; Lu et al. 2025; Yang et al. 2025). Simulation of verbal methods such as questionnaires, surveys and interviews was pursued for information about user thoughts, opinions, needs, experiences, actions or habits (Khanshan et al. 2024; Hämäläinen et al. 2023, 2022; Zhang et al. 2024b; Park et al. 2022; Ataei et al. 2025; Gu et al. 2025; Barambones et al. 2024; Salminen et al. 2024). Simulated feedback was also studied in the form of comments to videos and writing (Hung et al. 2025; Benharrak et al. 2024). Additionally, Al et al. (2024) compared topics generated from surveys by LLM and to crowd analysis, although crowdsourcing was to provide inadequate baseline, given the scenario's demand for expertise.

In traditional user-centered design, personas are fictional characters – storytelling devices used to narratively grasp user needs and behaviors (Lazik et al. 2025). Personas generated by LLMs have the potential to evoke empathy with the audience, but also detract from it by undermining in-depth comprehension by practitioners (Schuller et al. 2024; Sun et al. 2025b; Shin et al. 2024; Lazik et al. 2025; Sethi et al. 2025; Sattele and Ortiz 2024; Venkit et al. 2025). Basis in user data, enhanced further through classification by human experts can contribute to personas that are more plausible and complete (Jung et al. 2025; Shin et al. 2024; Kaate et al. 2025). Personas have also emerged as models applied in simulation (Gu et al. 2025).

For their capacity for advanced conversational interaction, LLM conversational agents were also studied in imitation games such as the Turing test (Gupta et al. 2025), and in virtual environments where they engaged in complex social simulations (Park et al. 2023). Although LLMs underperform task-specific models in Task-oriented dialogue (ToD) systems that assist users with performing tasks (Algherairy and Ahmed 2025; Hu et al. 2023), they can simulate users to train dialogue policy. Similar application was found in evaluation of interactive question answer (IQA) systems (Li et al. 2024c). Assuming that LLMs encode user experiences, Ma et al. (2025b) assessed them as a source of synthetic crowdsourced design.

4.1.4 Computer science and engineering

The motivation to study synthetic participants in technical fields was often to combat data scarcity and provide better quality software that delivers more useful information, such as more accurate recommendations or search results. In recommender systems, synthetic data was investigated as a complement to real behavioral data (Wang et al. 2025b) and to address the cold start problem (Huang et al. 2025b). Conversational recommender systems (CRS) represent an intuitive use case for simulating users who are seeking advice through dialogue (Zhao et al. 2025; Yoon et al. 2024; Zhu et al. 2025). In information retrieval, various studies have explored methods such as query and search-behavior generation, either outperforming traditional strategies, complementing them, or supporting their evaluation (Alaofi et al. 2023; Sannigrahi et al. 2024; Ran et al. 2025; Braga et al. 2024; Owoicho et al. 2023; Zhang et al. 2024a). LLM-generated word associations were compared to those created by humans by Bakaev et al. (2025) to understand differences in responses on computational linguistics level.

In the line of work related to Internet of Things (IoT), scarcity of human activity data from ambient sensors was tackled through simulated data to improve predictions for household resident behaviors, security anomalies (Leng et al. 2025; Xu et al. 2025; Yonekura et al. 2024), as well as strategies to support explainability (Fiori et al. 2024).

Steinmacher et al. (2024) also investigated whether LLMs can replicate the results of software engineering surveys targeting industry professionals or students.

4.1.5 Education

In education, understanding and modeling of students' grasp of curriculum can contribute to better serving of educational content (Lang 2023). LLMs were used to simulate students in writing essays (Herbold et al. 2023), coding (MacNeil et al. 2024; Zhan et al. 2025), and answering exam questions (Benedetto et al. 2024). In the context of knowledge measurement and item response theory, students were simulated to study psychometric alignment with real students and the potential for item validation (He-Yueya et al. 2024; Liu et al. 2025). Tabular student data for learning analytics was generated by Khalil et al. (2025) by using LLMs trained on real data. Conversational question answering and VR training systems simulated teachers, students and audiences in public speaking to enhance the interactivity of learning (Abbasiantaeb et al. 2024; Min and Jeong 2024).

4.1.6 Healthcare

The scarcity of certain types of patient data caused by strict privacy protections boosts the enthusiasm toward silicon participants for data-driven research in the healthcare domain. The algorithmic fidelity of synthetic data was studied for patient interviews and reporting (Amirova et al. 2024; Mori et al. 2024). To foster the abilities and confidence of learners in providing counseling, patient dialogues were generated synthetically (De et al. 2025; Gray et al. 2024; Wang et al. 2024). Further use cases for synthetic user health data include electronic health records (Abdel-Khalek et al. 2025), test cases for software development (Anderson et al. 2024) and personas for interventions (Smrke et al. 2025). Additionally, Jalali and Akhavan (2024) compared LLM-driven and human analysis of intervention interview transcripts.

4.1.7 Mobility & Tourism

Mobility data – important for purposes like urban planning and disease control – represent a type of human data primed for synthesis because of privacy barriers and low response rates involved with travel diaries being burdening on participants (Bhandari et al. 2024; Tzachristas et al. 2025; Ju et al. 2025). Encoding of travel experiences in LLM was studied in survey simulation (Ivasciuc et al. 2025), with a fruitful area of investigation focused on ethical concerns of falsified data, their analytical validity and reliability limitations linked to LLM-generated data (Sop and Kurcer 2024; Homer 2025).

4.1.8 Marketing

Driven by the need to understand demand, market research finds interest in the reduction of field work costs, increased research scale and number of variables to study. Some studies explored replication of hypotheses about consumer decision-making based on textual (Bickley et al. 2025) and cross-modal stimuli (Imschloss et al. 2025). Generated customer reviews with consideration for store category were studied by Shan and Jia (2024). Li et al. (2024b) used LLMs to evaluate perception of brands. Conjoint analysis was simulated as a source of information on preferences and willingness to pay (Brand et al. 2024). Beyond the naive approach of sheer simulation, Wang et al. (2025c) applied transfer learning with real data to reach better alignment of synthetic labels with reality.

4.1.9 Economics

Synthetic participants have also gained traction in studies focused on economic decision-making of humans. This includes replication of a Japanese economic survey (Kitadai et al. 2024), involving – among others – preferences, spending experiences and financial literacy. Alonso et al. (2025) studied hypothetical attitudes of various groups toward Central Bank Digital Currencies. Economic games provided the background for the study of LLMs' capacity for behavior tied to concepts like trust and strategic reasoning (Xie et al. 2024; Gao et al. 2025).

4.2 Task (RQ2)

RQ2: What types of task scenarios are synthetic participants applied to?

Depending on procedural or outcome-oriented focus, the surveyed simulation tasks are assigned into eight categories. For a transparent overview and definitions, they are presented in Table 6. A subset of works (38%) combined multiple tasks, most commonly generating personas to simulate other kinds of data (for complete distribution, see Figure 3).

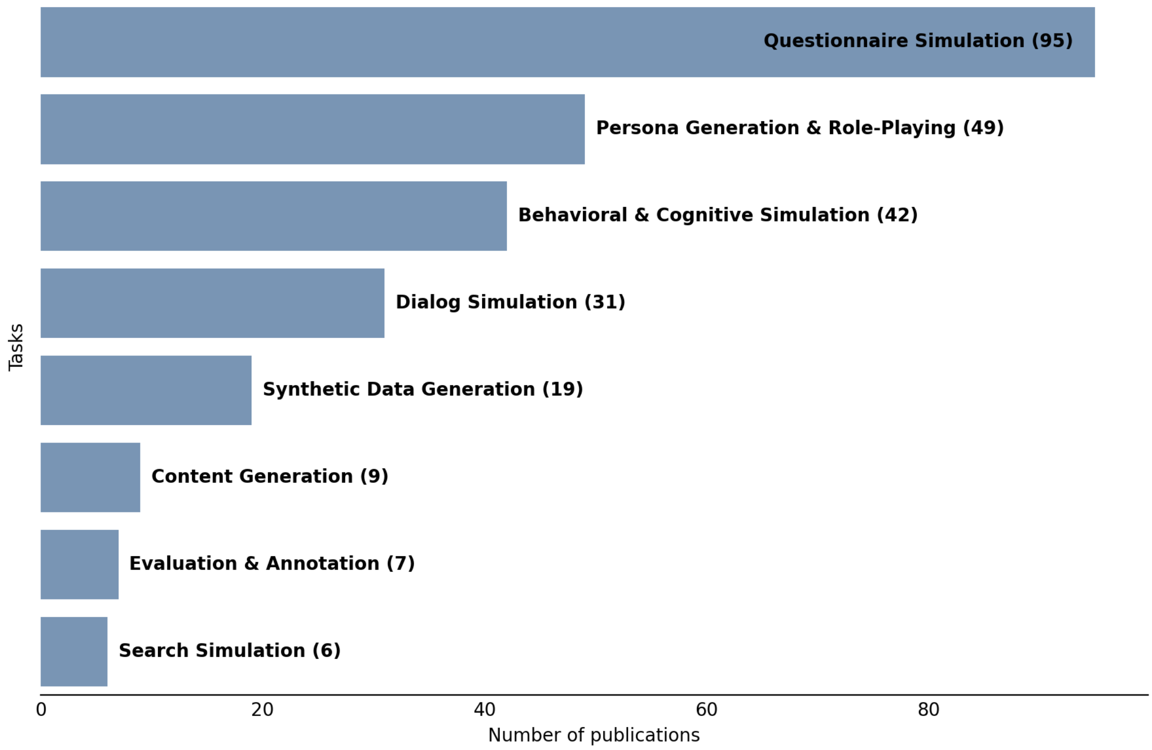

Fig. 3 Task category distribution in LLM-driven participant simulation.

To address these tasks, the response format generated by LLMs was most commonly free-form text (88), followed by categorical responses (66), scales such Likert (59), dialogue (29), numeric responses (26), list of values (26), probability distributions (13) and source code (5). A mismatch between the expected format and LLM output was sometimes encountered due to some models' inability to follow instructions (Almeida et al. 2024).

4.3 Issues (RQ3)

RQ3: What issues are associated with synthetic participants?

The surveyed literature demonstrates a variety of issues that hinder the algorithmic fidelity of LLM-based synthetic participants, evidenced by differences from real human subjects (Amirova et al. 2024; Argyle et al. 2023; Mori et al. 2024). While some studies provide valuable information by

Table 6 Task categories in LLM-driven participant simulation.

Task	Description	Publications
Questionnaire Simulation	Generation of answers to sets of various open- and close-ended questions; most encompass psychometric questionnaires and surveys regarding beliefs, opinions, experiences, etc.	Almeida et al. (2024); Alonso et al. (2025); Anderson et al. (2024); Argyle et al. (2023); Ball et al. (2025); Benedetto et al. (2024); Bernardelle et al. (2025); Bhandari et al. (2024, 2025); Bickley et al. (2025); Binz and Schulz (2023); Bisbee et al. (2024); Bodroza et al. (2024); Boelaert et al. (2025); Brand et al. (2024); Cava and Tagarelli (2025); Cho et al. (2024); Dey et al. (2025); Dominguez-Olmedo et al. (2024); Dorner et al. (2023); Durmus et al. (2024); Elyoseph et al. (2023); Giorgi et al. (2024); Gmyrek et al. (2025); Gupta et al. (2025); Hämäläinen et al. (2023); Hartmann et al. (2023); Havaei et al. (2024); He-Yueya et al. (2024); Heyde et al. (2025); Heyman and Heyman (2024); Huang et al. (2024a,b,c,d, 2025a); Hwang et al. (2023); Imschloss et al. (2025); Ivasciuc et al. (2025); Jiang et al. (2023, 2024); Jiaqi et al. (2025); Kalinin (2023); Kambhatla et al. (2025); Kang et al. (2025); Kim and Lee (2024); Kovač et al. (2024); Kwok et al. (2024); Lee et al. (2024a,b, 2025); Li et al. (2024b, 2025); Li and Qi (2025); Lim et al. (2025); Liu et al. (2025); Ma et al. (2024, 2025a); Molchanova et al. (2025); Motoki et al. (2024); Neumann et al. (2025); Niszczota et al. (2025); Park and Kim (2024); Park et al. (2024a,b); Pellert et al. (2024); Petrov et al. (2024); Qi et al. (2025); Qu and Wang (2024); Robinson et al. (2023); Rozado (2023); Rutinowski et al. (2024); Salecha et al. (2024); Sanders et al. (2023); Santurkar et al. (2023); Serapio-García et al. (2025); Shrestha et al. (2024); Sop and Kurcer (2024); Steinmacher et al. (2024); Suguri Motoki et al. (2024); Suh et al. (2025); Sun et al. (2024); Tavast et al. (2022); Tjuatja et al. (2024); Trott (2024); Tzachristas et al. (2025); Venkit et al. (2025); Wang et al. (2025a,c,d,e); Winter et al. (2024); Zhang et al. (2024c, 2025); Zhou et al. (2025)
Persona Generation & Role-Playing	LLM automating or assisting with the generation of vivid personas and their characteristics, including demographics and personality. Intended for practitioners or for synthetic role-playing to obtain simulated behaviors and responses.	Amirova et al. (2024); Ataei et al. (2025); Bakaev et al. (2025); Barambones et al. (2024); Benharrak et al. (2024); Bernardelle et al. (2025); Bisbee et al. (2024); Cava and Tagarelli (2025); Cheng et al. (2023); Chuang et al. (2024); De et al. (2025); Dey et al. (2025); Giorgi et al. (2024); Gray et al. (2024); Gu et al. (2025); Huang et al. (2024d); Jiang et al. (2024); Ju et al. (2025); Jung et al. (2025); Kaate et al. (2025); Kang et al. (2025); Kapania et al. (2025); Lazik et al. (2025); Lee et al. (2024b); Leng et al. (2025); Li et al. (2025); Lo et al. (2025); Lu et al. (2025); Ma et al. (2024); Milička et al. (2024); Min and Jeong (2024); Nóbrega et al. (2025); Park et al. (2022); Salminen et al. (2024); Sattele and Ortiz (2024); Schuller et al. (2024); Sethi et al. (2025); Shin et al. (2024); Smrke et al. (2025); Suguri Motoki et al. (2024); Sun et al. (2025b); Tzachristas et al. (2025); Venkit et al. (2025); Wang et al. (2024); Winter et al. (2024); Xiang et al. (2024); Yang et al. (2025); Zhao et al. (2025); Zhu et al. (2025)
Behavioral & Cognitive Simulation	Replication of the outcomes of human internal cognition and external behavior, experimentally assessed against effects, patterns and biases observed in humans, or simulated in sandbox environments.	Aher et al. (2023); Almeida et al. (2024); Alonso et al. (2025); Amirizaniani et al. (2024); Bakaev et al. (2025); Benedetto et al. (2024); Binz and Schulz (2023); Chuang et al. (2024); Du et al. (2024); Flamino et al. (2025); Gao et al. (2025); Griffin et al. (2023); Gupta et al. (2025); Hagendorff et al. (2023); Havaei et al. (2024); Homer (2025); Huang et al. (2025b); Huff and Ulakçı (2024); Imschloss et al. (2025); Ju et al. (2025); Khanshan et al. (2024); Kitadai et al. (2024); Leng et al. (2025); Li et al. (2024a); Lo et al. (2025); Lu et al. (2025); Manning et al. (2024); Marjieh et al. (2024); Milička et al. (2024); Owoicho et al. (2023); Park et al. (2022, 2023, 2024a); Wang et al. (2025b); Xiang et al. (2024); Xie et al. (2024); Xu et al. (2025); Yonekura et al. (2024); Yoon et al. (2024); Zhan et al. (2025); Zhang et al. (2024a,d)
Dialog Simulation	Simulation of back-and-forth conversations as human-agent social interactions, including interviews and question answering.	Abbasiantaeb et al. (2024); Algherairy and Ahmed (2025); Amirova et al. (2024); Ataei et al. (2025); Barambones et al. (2024); Chuang et al. (2024); De et al. (2025); Du et al. (2024); Fang et al. (2024); Flamino et al. (2025); Gray et al. (2024); Gu et al. (2025); Hämäläinen et al. (2022); Han et al. (2024); Kaate et al. (2025); Kapania et al. (2025); Li et al. (2024c); Matsumura et al. (2025); Min and Jeong (2024); Owoicho et al. (2023); Park et al. (2023); Shin et al. (2024); Sparrenberg et al. (2024); Sun et al. (2025b); Wang et al. (2024); Yang et al. (2025); Yoon et al. (2024); Zhang et al. (2024b,d); Zhao et al. (2025); Zhu et al. (2025)
Synthetic Data Generation	Generation of datasets of tabular, behavioral or textual data and records that capture a target population.	Abdel-Khalek et al. (2025); Al et al. (2024); Alaofi et al. (2023); Anderson et al. (2024); Bhandari et al. (2024); Braga et al. (2024); Busker et al. (2025); Han et al. (2024); Jiang et al. (2025); Khalil et al. (2025); Lee et al. (2025); Leng et al. (2025); MacNeil et al. (2024); Mori et al. (2024); Sannigrahi et al. (2024); Sop and Kurcer (2024); Suguri Motoki et al. (2024); Tzachristas et al. (2025); Xu et al. (2025)
Content Generation	Creation of text-based content, including ideas, reviews, essays and source code.	Amirizaniani et al. (2024); Benharrak et al. (2024); Herbold et al. (2023); Hung et al. (2025); Khanshan et al. (2024); Ma et al. (2025b); MacNeil et al. (2024); Shan and Jia (2024); Zhan et al. (2025)
Evaluation & Annotation	Simulation of rating and labeling that typically depend on human reasoning and subjective judgement to analyze qualitative data.	Al et al. (2024); Fiori et al. (2024); Giorgi et al. (2024); Hu et al. (2023); Jalali and Akhavan (2024); Li et al. (2024c); Sun et al. (2025a)
Search Simulation	Simulation of information needs expressed through queries and information behaviors in interactive search systems.	Alaofi et al. (2023); Owoicho et al. (2023); Ran et al. (2025); Sannigrahi et al. (2024); Wang et al. (2025b); Zhang et al. (2024a)

assessing reliability or construct validity (Alonso et al. 2025; Dorner et al. 2023), such measures may characterize only the AI model, without generalizing well to humans (Benedetto et al. 2024). Assuming inherent alignment with humans (or LLM simulation could be bias-free (Jalali and Akhavan 2024)), based on no evaluation, or solely on pattern matches that are isolated or incomplete, could lead to misleading conclusions. Therefore, this section comprehensively analyzes the issues documented in empirical findings.

4.3.1 Misalignment with cognitive processes

With algorithmic fidelity as the measure of success for generated data, studies in psychology become uniquely relevant. Their perspective reaches beyond descriptive comparison between observational ground truth and simulation at the level of behavioral outcomes. Beyond looking at outcomes in a vacuum, it provides comprehensive conceptual framework to systematically explain the nature of disparities between humans and the LLMs through the understanding of latent psychological and neurophysiological processes. As factors that affect perceptions and behaviors, these have wide-spanning downstream implications on the usefulness and reliability of synthetic participants crossing into other fields.

The differences between human intelligence and LLMs are rooted in their contrasting nature and function. Humans are embodied beings that process and act upon sensory input to survive and reproduce. LLMs are predictors of plausible text (Gao et al. 2025), thus lacking a life experience (Homer 2025). Some publications acknowledge the inherent differences, stressing that linguistic skills cannot be conflated with genuine reasoning and understanding (Pellert et al. 2024; Gu et al. 2025), but usually offering limited extent and detail (Bickley et al. 2025). Others exaggerate LLMs capabilities by ascribing them human-level intelligence and other anthropomorphic traits (Wang et al. 2025b; Zhu et al. 2025; Shin et al. 2024) or minimize the subjectivity of human cognition, such as when it concerns processing of factual information (Benedetto et al. 2024). Our analysis addresses this lack of systematic grasp on the distinction by isolating its key psychological factors. As we show below, LLMs are inadequate at simulating the complexity of aspects that define humans as people: thoughts and reasoning (causal, affective, moral, high), decision-making, cognitive abilities, emotions and personalities.

With improved problem-solving being a touted feature of newer LLMs, GPT-4's outperformance of humans in cognitive reflection tests can be seen as a natural outcome of tuning on cognitive reasoning benchmarks (Hagendorff et al. 2023). GPT-3 as an earlier model showed cognitive errors at rates similar to humans, raising its potential for participant simulation. However, it still lacks the varied biases and patterns that humans demonstrate during decision-making and deliberation (e.g., no reflection effect), as well as falls short in causal reasoning (Binz and Schulz 2023). LLMs also achieve mixed results in moral and legal reasoning (e.g., attributing intentionality to actions), with the best model (GPT-4) still showing systematic differences from humans (Almeida et al. 2024). Although LLMs lack human-like memory, Huff and Ulakçı (2024) showed their promise in predicting memorability as a factor of recall.

Emotional awareness assessments demonstrated an ability to attribute potential relevant emotional reactions to self and others – to an unrealistically encyclopedic degree of almost perfect test scores (Elyoseph et al. 2023). This pairs with high projected emotional intelligence (Huang et al. 2024c). However, while emotional responses selected by the LLMs can be deemed socially appropriate, they deviate from human responses (e.g., high emotionality as the default state, overexaggeration, reduction or erasure of negative emotions) (Huang et al. 2024b; Park and Kim 2024; Tavast et al. 2022). Higher reasoning investigated through the Theory of Mind shows LLMs as inferior to human reasoning, even after improvements by modeling intentions and emotions (Amirizaniani et al. 2024). The authors attributed this to lack of subjectivity in reasoning.

Personality traits projected by LLMs diverge between models and prompting approaches (Bhandari et al. 2025; Niszczota et al. 2025; Li and Qi 2025; Huang et al. 2024c; Pellert et al. 2024). However, the models' lack of latent personality prevents the use of traditional psychometric frameworks for valid model-to-model comparison, as evidenced by a lack of consistency (e.g., positive answers to reverse-coded items) and separation of features (Dorner et al. 2023). Although models can be steered with prompts, outputs tend occupy an area of the personality space bounded by or in the proximity to the model's base distribution (Huang et al. 2024a). Larger and instruction-tuned models adopt imposed personality with higher accuracy (Serapio-García et al. 2025) than smaller models (Cava and Tagarelli

2025). Including personality traits in prompts can ensure higher consistency in personality assessment questionnaires (Huang et al. 2024d; Lo et al. 2025; Sparrenberg et al. 2024), albeit unrealistically high and more factorially pure compared to humans who are more multifaceted (Wang et al. 2025e).

Simulated personalities also deviate from humans in their decisions (Huang et al. 2024d). Furthermore, issues in text generated based on personality include exaggerating stereotypes, logical inconsistencies (changed motivation, becoming a different person) and lack of meaning and perceived realism (Jiang et al. 2024; Molchanova et al. 2025; Han et al. 2024). The existence of cognitive, social and cultural mediators and individual differences makes realistic personality simulation challenging (Lim et al. 2025; Wang et al. 2025d; Dey et al. 2025).

This can have negative downstream effects on various pursuits dependent on psychological and behavioral accuracy to inform decisions or guide automation. A small selection of areas affected downstream includes simulated decision-making in the context of services, risk-taking or moral dilemmas (Bickley et al. 2025; Binz and Schulz 2023; Kovač et al. 2024; Huang et al. 2024d), analytical and economic reasoning (Homer 2025; Jalali and Akhavan 2024; Gao et al. 2025), problem-solving in the context of programming or writing queries (MacNeil et al. 2024; Alaofi et al. 2023) or using personality to model behaviors in focus groups or travel flow (Zhang et al. 2024b; Ju et al. 2025).

4.3.2 Distortions and biases

Validations of LLM-simulated participants across a variety of contexts indicate mixed results in alignment with human data (Huang et al. 2024c; Xiang et al. 2024; Ma et al. 2025b; Lazik et al. 2025). In the full corpus covered by this review, there was no domain or task as a clear boundary where LLM simulation performs better or worse. Research that did not provide direct comparison to humans also tended to discuss the potential of machine biases (Benedetto et al. 2024).

Systematic gaps in algorithmic fidelity were formally described by Aher et al. (2023) as distortions. Distortion is distinct from bias, which human data is also subject to naturally (e.g., cognitive bias) as the bias itself potentially captures important psychological, social or behavioral effects. Rather, distortion represents disparity in bias between humans and the simulation. Various distortions may overlap to affect replication of human patterns (e.g., no central tendency bias) (Aher et al. 2023; Park and Kim 2024; Yoon et al. 2024; Binz and Schulz 2023), unrepresentatively reduce or overexaggerate their magnitude (Almeida et al. 2024; Aher et al. 2023; Wang et al. 2025e; Jiang et al. 2024; Huang et al. 2024b; Elyoseph et al. 2023), or introduce altogether new machine patterns (e.g., conflating the best answer as the most probable one; always selecting multiple choices in multichoice questions (Robinson et al. 2023; Kitadai et al. 2024)).

Although the exact sources of distortions are difficult to pinpoint since LLMs are black boxes with a multitude of factors involved in their training, they have two main points of origin: the training data, and the methods used to train and fine-tune the model (such as Reinforcement Learning from Human Feedback, i.e. RLHF, a guardrail feature that makes LLM outputs usable, aligning them with human needs and expectations) (Lu et al. 2025; Salminen et al. 2024; Sattele and Ortiz 2024). Below, we list the (often overlapping) types of distortions that manifest in LLM-generated participant data.

Low depth and variability. Perhaps the most universal and ubiquitous bias in synthetic participant data is the lack of realistic variability and diversity (Zhang et al. 2025; Hämäläinen et al. 2023; Zhang et al. 2024b; Ma et al. 2025b; Barambones et al. 2024; Alaofi et al. 2023; Liu et al. 2025; Almeida et al. 2024; Yoon et al. 2024; Lazik et al. 2025; Bakaev et al. 2025). This can be explained by LLMs not possessing genuine experiences of ordinary constantly-changing life as the basis for authentic contextually-aware narratives (Bakaev et al. 2025; Venkit et al. 2025; Xu et al. 2025). As proficient text generators, LLM responses can have deceptively high depth: they are verbose (Benharrak et al. 2024; Abbasiantaeb et al. 2024), their text tends to be overstructured (Herbold et al. 2023; Amirova et al. 2024) grammatically correct (Gupta et al. 2025), and with high lexical diversity (Algherairy and Ahmed 2025). However, they are also shallow, non-specific, repetitive and homogeneously perspectiveless, lacking in nuance and contextual awareness (Amirizaniani et al. 2024; Benharrak et al. 2024; Homer 2025) while exaggerating some effects found in humans (Almeida et al. 2024). Without personal preferences, requests that LLMs make are generic (Yoon et al. 2024). It does not encourage in-depth explorative proactivity (Shin et al. 2024; Gu et al. 2025). LLMs systematically fail to infer implications that are intuitive to human reasoning, even from simple instructions (Gao et al. 2025; Homer 2025; Park et al. 2023) or image contents, despite multimodal LLMs being capable to to identify image contents on a descriptive level (Imschloss et al. 2025). On the syntactic surface, outputs may seem elaborate, yet remain narratively reductive (Venkit et al. 2025), potentially bordering on incoherent rambling (Hämäläinen et al. 2023).

Creativity as a source of variability in human responses is tied to cognitive processes. Humans provide in significantly more original responses than LLMs, including errors caused by cognitive abilities (e.g., memory lapses about specific facts), misunderstanding of the question, with structure that follows a trail of reasoning instead of a rigid structure, and sharing of potentially less relevant details (Gupta et al. 2025). Humans are also more inclined to creative exploration, as they can find more topics of discussion even after multiple iterations (Zhang et al. 2024b). Generated datasets may compete with a single participant, but depending on the task, two or more participants comprise a better sample (Trott 2024).

In practical applications dependent on data variability, simulated data may have neutral or even negative effects on method performance (Leng et al. 2025). Examples that help conceptualize the level the variability of LLM-generated participant data include heuristic idea generation by designers using personas (Gu et al. 2025), and test-retest of the typicality rating task, where it resembled the same person between consecutive days (Heyman and Heyman 2024). Additionally, in a simple rating task (e.g., measuring perceived truthfulness of statements (Griffin et al. 2023)), silicon participants may inflate variability instead.

Inconsistency. The lack of narrative depth also negatively impacts internal consistency (Jiang et al. 2024). Although generated data such as personas may outwardly project high consistency (Salminen et al. 2024), this is likely because it is moderated by machine biases, including low diversity, stereotypicality and positivity (Lazik et al. 2025; Venkit et al. 2025). Inconsistencies in LLM reasoning are apparent, such as justifying an option by risk aversion, but selecting the same option regardless of risk (Gao et al. 2025). Paradoxically, natural contradictions which are rooted in the plurality of lived human experiences were found missing from LLM simulations (Venkit et al. 2025). For example, the LLM-generated persona of a social worker with shopping addiction is inconsistent from real data due to financial constraints (Salminen et al. 2024), while also lacking the grounded nuance explaining how the addiction developed in such a person and how it affects them. Furthermore, as conversations become longer, expanded context and elapsed time increase inconsistency, even enabling extraneous manipulation (Kovač et al. 2024; Park and Kim 2024). Therefore, the assumption of treating LLM conversations as if they represented separate individuals may itself be misguided (Kovač et al. 2024; Park and Kim 2024).

Divergence from human patterns. In a variety of cases, statistical mismatch between LLM-simulated and human data indicated no replication of factually-supported effects and patterns. This includes cognitive biases (Binz and Schulz 2023), opinions on social issues (Durmus et al. 2024), and generated personas based on surveys (Shin et al. 2024). Even when provided with relevant personas, established characteristics of human participants may not be fully captured (He-Yueya et al. 2024; Li and Qi 2025; Sun et al. 2025b), including complex approaches such as culturally-informed personality assessment (Dey et al. 2025). Realistically simulating complex multi-participant interactions like focus groups may also be out of scope for LLMs, considering the lack of group dynamics, disagreement and differentiation between individual perspectives (Zhang et al. 2024b).

Latent Trait Manipulation Dilemma. Given that LLMs lack deeply represented latent traits, there is the dilemma that any simulation in tasks affected by latent constructs or individual biases is predicated on prompt manipulation. In psychometric evaluation, this can be reflected by distribution mismatch (e.g., too narrow for proficiency (Liu et al. 2025), too broad for created bugs (MacNeil et al. 2024)), as well as lack of construct validity. Dorner et al. (2023) showed this through poor model fit and warned against misinterpreting α as high reliability, given the violation of good fit assumption. To some extent, better alignment with latent traits may be achieved through prompt manipulation with relevant (Petrov et al. 2024) and explicit (Giorgi et al. 2024) information. However, this is unreliable as LLM outputs exhibit significant prompt brittleness, significantly different from the effects that wording has on humans (Tjuatja et al. 2024; De et al. 2025). They are sensitive to order, language, as well as minor word changes (Barambones et al. 2024; Gao et al. 2025). Therefore, more complex prompts may also be subject to more unpredictable outputs due to distortion.

Deficient representativity. Silicon participants lack representativity (Benharrak et al. 2024; Venkit et al. 2025; Anderson et al. 2024). Training datasets of LLMs have inherent racial and gender biases, being generated overwhelmingly by white males (Aher et al. 2023; Huang et al. 2024c; Barambones et al. 2024; Pellert et al. 2024). They are also USA-centric, with discrepancies that indicate missing social and cultural nuances of other countries (Kitadai et al. 2024). Even while processing actual human data, LLMs still overrepresent certain groups over others (Jung et al. 2025). The lack of representativity is not purely demographic, as artifacts can appear surrounding specific topics. For example, Hämäläinen et al. (2023) discovered a "Journey bias". Synthetic participants, when asked to name an artistic video game, overwhelmingly picked Journey – an over-a-decade-old darling of critical and online discourse about games as art, and therefore more likely to be over-represented by LLMs. By comparison, human responses were significantly more varied and showed recency bias.

Stereotypes. The tendency to rely on stereotypes is a consistent pattern across various studies on LLM synthetic responses (Durmus et al. 2024; Kwok et al. 2024; Ma et al. 2024; Benharrak et al. 2024; Lazik et al. 2025), demonstrating the need to use systematic measurement methods such as CoMPosT (Sun et al. 2025b). In-depth analysis by Venkit et al. (2025) also identified the presence of adjacent phenomena, including exoticism, dehumanization, erasure and benevolent bias. Instead of representing minorities within the depth of human experience, LLMs cast minorities into flat archetypal roles of "cultural ambassadors" and "resilient survivors".

Helpfulness, harmlessness, honesty (HHH). Biases linked to the attunement of LLM outputs toward usefulness are linked to RLHF (Sparrenberg et al. 2024; Dorner et al. 2023; Li and Qi 2025; Kwok et al. 2024), an approach driven by user preferences. LLM responses inherit a social desirability bias (Salecha et al. 2024; Molchanova et al. 2025; Li and Qi 2025; Petrov et al. 2024; Huang et al. 2024c), emphasizing traits like friendliness, extraversion, conscientiousness and emotional stability. Helpfulness can extend to decisions that systematically diverge from humans, such as over-leaning on fairness and prosocial behavior toward game opponents (Gao et al. 2025; Xie et al. 2024). Positivity bias can be mirrored in favorable reviews (Shan and Jia 2024), flattering and benevolent personas based on positive stereotypes (Lazik et al. 2025; Sattele and Ortiz 2024), high tolerance against issues (Xiang et al. 2024), or overtly flexible preferences that conflate having dislikes with pickiness while obfuscating latent reasons (Yoon et al. 2024). Agree bias means they tend to be sycophantic (Dorner et al. 2023). They adhere to a formal conversational style (Park et al. 2023), which may appear milquetoast in simulation of online troll behavior, which is commonly toxic and inappropriate (Park et al. 2023). Emotional baselines are exaggerated, although negative emotions are reduced, or avoided completely (e.g., jealousy) (Huang et al. 2024b).

Temporal lock-in. Human thoughts, opinions and behaviors are in constant flux. LLMs struggle at capturing social change, entrenching the most dominant values over time as a result of their training on past data (Lazik et al. 2025; Benharrak et al. 2024). For retrodiction, specifying the year in the prompt might increase accuracy (Li et al. 2024b), although generalization to nuanced out-of-distribution inputs (as opposed to, e.g., public perceptions of established car brands) is without a basis.

Hyperaccuracy. The inclination of LLMs toward perfectly memorized, accurate, detailed and highly structured answers is helpful for question-answering systems. What is projected as better knowledge, and reasoning does not plausibly reflect actual human behavior and reasoning (Elyoseph et al. 2023; Milička et al. 2024; Aher et al. 2023; Benedetto et al. 2024; Amirova et al. 2024), for example by citing guidelines instead of reflecting beliefs from human interviews (Amirova et al. 2024). With larger RLHF-tuned models like GPT-4, the issue is becoming progressively worse (Aher et al. 2023; Benedetto et al. 2024). LLMs may also still fail to grasp specific domain knowledge (Shin et al. 2024).

Hallucinations. As an inherent property of LLMs (Xiang et al. 2024; Hämäläinen et al. 2023; Sun et al. 2025b; Braga et al. 2024), hallucinations may not refer only to factual errors (Sattele and Ortiz 2024), but also to the models making unpredictable assumptions (Wang et al. 2025c), or drifting away from the persona, context or task they were given, resulting in fabricating facts and rambling (Kaate et al. 2025; Homer 2025). In a fashion similar to hyperaccuracy, LLMs also generate a lot more topics (e.g., needs, questions, usability issues) that are not supported by human responses. As potential hallucinations lacking full contextual awareness, these could be misleading and obscure the most pertinent real answers (Xiang et al. 2024; Hämäläinen et al. 2023; Zhang et al. 2024b; Ataei et al. 2025; Hung et al. 2025; Abbasiantaeb et al. 2024). In spite of this concern, users of synthetic feedback may feel pressured to incorporate it (Benharrak et al. 2024).

4.3.3 Misleading believability

In contrast to lackluster algorithmic fidelity, the believability of LLM participants can be excellent (Park et al. 2022; Schuller et al. 2024; Barambones et al. 2024; Venkit et al. 2025; Wang et al. 2025b; Gray et al. 2024). Judging by appearance, synthetic responses can be difficult to distinguish (Schuller et al. 2024), their plausibility further reinforced by linguistic elaborateness and the ability to adapt to prompts (Milička et al. 2024; Benedetto et al. 2024; De et al. 2025). Supported by some degree of quantitative and qualitative similarity (statistically significant effects, correlations, linguistic and thematic patterns (Aher et al. 2023; Li et al. 2024c; Wang et al. 2025b; Fiori et al. 2024; Liu et al. 2025; De et al. 2025)), they inspire confidence toward the possibility of inferring valuable insights (Benharrak et al. 2024; Zhao et al. 2025; Zhu et al. 2025; Li et al. 2024b; Xie et al. 2024).

However, believability is superficial. Although even experts may subjectively find synthetic personas indistinguishable from human-generated ones (Schuller et al. 2024), deeper investigation reveals significant differences in realism and stereotypicality (Lazik et al. 2025). As LLMs become more familiar, people also grow more proficient at identifying LLM imitations based on creative and structural factors (Gupta et al. 2025), which is a contrast to prior findings (Park et al. 2022; Hämäläinen et al. 2023), although some may still associate non-indicative factors like grammatical correctness with human likeness (Lazik et al. 2025). Believability may actually be more detrimental than beneficial by lending false credibility to misleading conclusions based on hallucinations and lack of nuance (Hung et al. 2025; Wang et al. 2025c; Hämäläinen et al. 2023). This cautious sentiment is echoed across a significant portion of primary studies, which stress that synthetic participants should not be viewed as substitutes for human participants (Ma et al. 2024; Mori et al. 2024; Almeida et al. 2024; Boelaert et al. 2025; Lu et al. 2025; Park et al. 2024b; Zhang et al. 2024b; Park et al. 2022; Brand et al. 2024; Heyman and Heyman 2024), but as a complementary concept.

Trust in LLM-generated personas may also impede rather than support practitioners' ability to empathize and understand others by reducing the depth of mental engagement and connection (Shin et al. 2024; Gu et al. 2025; Schuller et al. 2024). Approaches that emphasize augmentation over full automation as part of the design of faithful silicon participants are underrepresented.

High believability also highlights the risks of LLMs as tools for producing fraudulent data that does not meet quality expectations and distorts patterns and relationships (Sop and Kurcer 2024; Lu et al. 2025). Participants and researchers may contaminate datasets with believable data that may subvert traditional quality checks, especially for responses that are not text-based (Heyman and Heyman 2024).

LLMs themselves can also overestimate their own proficiency at simulating participants. They have a self-favoring bias (Li et al. 2024c), given that the text they generate is inherently an output of their internal embeddings. Wang et al. (2024) demonstrated that an LLM evaluator preferred conversations by baseline GPT-4 over those of the proposed framework, which was preferred by experts and users. Self-supervised learning may thus propagate and reinforce such errors.

4.3.4 Overfitting and contamination

If a study as a source of ground truth about participants was already published online, there exists a likelihood that some information describing tasks and data was already used to train LLMs. Access to this information may inflate algorithmic fidelity for in-distribution inputs without effective generalization to out-of-distribution inputs, such as altered tasks or different populations (Gao et al. 2025; Alonso et al. 2025; Li et al. 2024b; Steinmacher et al. 2024; Benedetto et al. 2024; Ivasciuc et al. 2025). This is a concern regardless of the scale of present distortions. Some studies partially address this by focusing on ground truth that was only published recently and is thus less likely to be a part of the LLMs dataset (Benedetto et al. 2024). Nonetheless, using datasets unseen prior (e.g., Romanian gen-Z travel experiences by Ivasciuc et al. (2025)) is a more reliable approach to mitigate contamination.

For LLMs fine-tuned with contextual information, contexts not included during fine-tuning could have still been seen by the model during standard pre-training (Bhandari et al. 2024; Huang et al. 2025b). This complicates empirical evaluation of external validity, including application to different domains and faithful portrayal of a cold start (Bhandari et al. 2024; Huang et al. 2025b).

Benchmark overfitting presents a particular subproblem as models are tuned to project higher human likeness along with better cognitive and problem-solving capabilities. For example, LLMs may appear to apply Theory of Mind by memorizing correct solutions for benchmark experiments (Heyman and Heyman 2024). Future models may show increasing alignment with established experiments and psychometric inventories as a result (Aher et al. 2023; Pellert et al. 2024).

4.4 Prompt engineering (RQ4)

RQ4: What prompting strategies are used to generate synthetic participants?

The overwhelming majority of participant simulations used zero-shot prompts. This aligns with the ostensible prospect of zero-shot prompts to simulate diverse tasks based solely on their natural language definition, thereby navigating around the steep data requirements associated with training specialized models (e.g., multi-layer perceptrons) (Wang et al. 2025b). This is undermined by LLMs' inability to capture out-of-distribution behaviors (Gao et al. 2025). Few-shot prompting can be more accurate (Hu et al. 2023), although it still deviates from human data (Gao et al. 2025; Suh et al. 2025). Examples, whether they are one-shot (Alaofi et al. 2023; Barambones et al. 2024; Khanshan et al. 2024; Gray et al. 2024; Robinson et al. 2023; Schuller et al. 2024; Sun et al. 2025b) or multiple (Benharrak et al. 2024; Gao et al. 2025; Marjieh et al. 2024; Hämäläinen et al. 2023; He-Yueya et al. 2024; Hung et al. 2025; Kitadai et al. 2024; Leng et al. 2025; Ma et al. 2025b; Owoicho et al. 2023; Park et al. 2022; Sannigrahi et al. 2024; Suh et al. 2025; Wang et al. 2025b,c) still prove insufficient at compensating for lack of encoding of deeper contextually-relevant associations.

Mixed results are also achieved by directing LLMs to engage in chain-of-thought reasoning, inconsistently across models (Hagendorff et al. 2023; Xie et al. 2024). Generated rationales may not improve simulation accuracy or fidelity (Benedetto et al. 2024; Li and Qi 2025). Reasoning frameworks such as Belief-Desire-Intention modeling may yield better alignment with human patterns, although it does not solve inherent biases such as lack of variability and amplified prosocial preferences (Xie et al. 2024).

Prompt chaining decomposes tasks into subtasks by passing the output of one prompt as input to another (Hung et al. 2025; Salminen et al. 2024; Tavast et al. 2022). Iterative critiquing of previous outputs can improve their diversity, although not to a human level (Ma et al. 2025b). Retrieval-augmented generation (RAG) as a means to incorporate vast document-based context into prompts can support better human alignment, although differences and hallucinations persist (Gao et al. 2025; Kaate et al. 2025; Leng et al. 2025; Sun et al. 2025b; Dey et al. 2025).

Less commonly used strategies include ReAct (2) (Xiang et al. 2024; Zhang et al. 2024a), self-consistency (1) (Winter et al. 2024), meta prompting (1) (Almeida et al. 2024), and tree-of-thoughts (1) (Zhan et al. 2025). The full distribution of prompt engineering strategies is shown in Figure 4.

The challenges of prompt engineering are complex. High prompt sensitivity can unpredictably alter outcomes because of seemingly insignificant differences, such as the framing of the same response as therapy conversation or blog post (Benedetto et al. 2024; Suh et al. 2025; Mori et al. 2024). Prompts tuned for one model do not generalize well to other models (Benedetto et al. 2024; Wang et al. 2025b). Adding information and directives may improve algorithmic fidelity, but large prompts can overwhelm LLMs, causing key information to be ignored (Schuller et al. 2024). Complex token-heavy prompts can also significantly inflate costs, highlighting the need for efficiency through prompt compression that selects and prioritizes important information (Xu et al. 2025).

4.5 Participant modeling (RQ5)

RQ5: What participant modeling strategies are used in LLM generation of synthetic participant data?

As LLMs approach the limit of being trained on most available quality data, the algorithmic fidelity of synthetic participants cannot be expected to improve by itself, just through the introduction of bigger and better models (Bakaev et al. 2025; Gao et al. 2025). To elevate simulation, capturing the traits of the participants through prompt design, model-based frameworks and task-specific fine-tuning is essential.

Personas can be used to model participants for steering LLMs, increasing the algorithmic fidelity, depth and variability of simulation. Among the three persona-based approaches proposed by (Gerosa et al. 2024), single-persona was the most common, followed by mega-persona, then multi-persona after a significant gap (see Figure 5). By enabling higher granularity, the single-persona approach can be considered as the most valid, whereas delegating control of an entire simulated population or group to

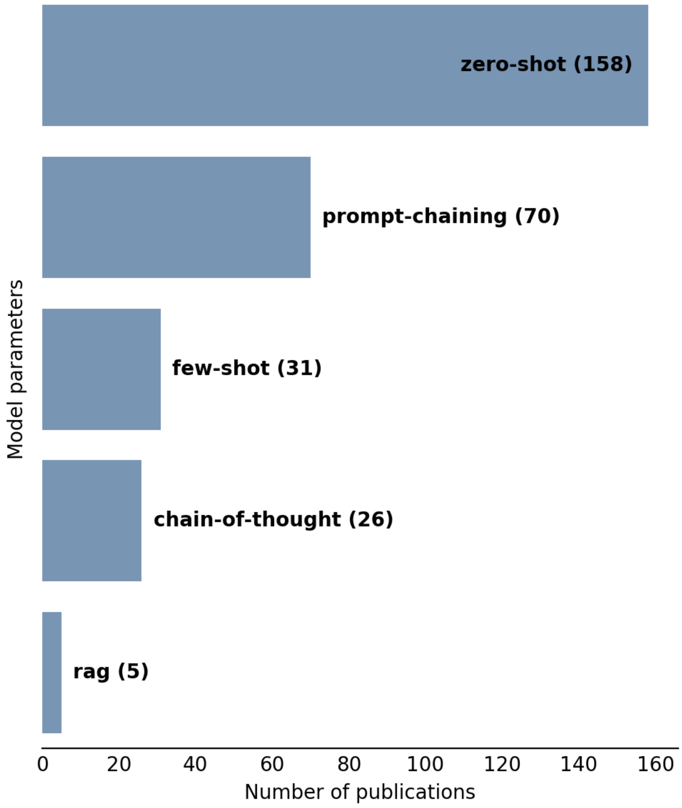

Fig. 4 Prompt engineering method distribution in LLM-driven participant simulation.

the LLM can result in data that does not fit established models and embellishes relationships (Sop and Kurcer 2024).

The central challenge of persona design lies in imparting the LLMs with suitable information that it can interpret to increase the realistic representation of latent factors. In tandem with LLMs' inability to deduce latent implications from instructions or images (Gao et al. 2025; Homer 2025; Imschloss et al. 2025), narratively fine-grained personas that provided more explicit and relevant information information performed best, whereas implicit and irrelevant information yielded neutral or negative effects (Giorgi et al. 2024; Wang et al. 2025e). Higher contextual volume does not inherently equate higher-quality personas, as evidenced by erratic behavior as input size increases (Park et al. 2023). Even small changes can have unpredictable consequences without validation (Tjuatja et al. 2024) (e.g. naming the persona (Aher et al. 2023)). With large context, LLMs tend to homogeneously combine all the information available (Barambones et al. 2024). Their inability to account for varying relevance of contextual details results in underwhelming performance in directions where simpler context is more effective (Zhu et al. 2025).

Demographics (e.g., age, gender, nationality, education, occupation), represent information most commonly modeled by personas, although their effectiveness in eliciting fidelity was often found mixed to questionable (Petrov et al. 2024; Anderson et al. 2024; Brand et al. 2024; Tzachristas et al. 2025; Kitadai et al. 2024; Wang et al. 2025e; Li and Qi 2025; Yang et al. 2025; Hung et al. 2025; Barambones et al. 2024; Xiang et al. 2024). The limited instances where demographics yielded positive results could arguably be attributed to encoding of associations that do not require deep nuance and understanding, such as physical and mental health's being influenced by age and country (Wang et al. 2025e), and daily travel habits influenced by marital status and the number of children in the household (Bhandari et al. 2024). Echoing the significance of explicitness and contextual relevance, personas were improved beyond generic demographics, such as through cultural context and verbal character descriptions in personality assessment (Dey et al. 2025; Petrov et al. 2024), and preferences in economic decision-making (Kitadai et al. 2024).

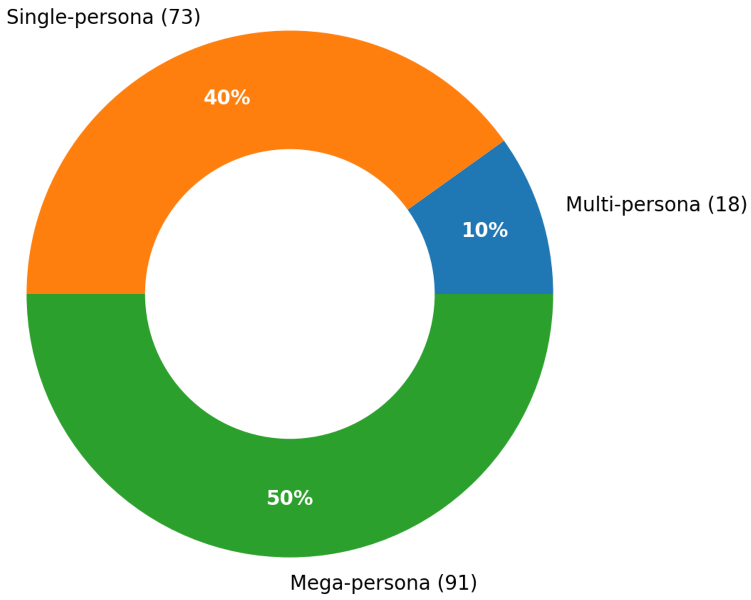

Fig. 5 Persona-based method distribution in LLM-driven participant simulation.

The dependence between persona explicitness and context-relevance, and their effectiveness in generating valid data, is best illustrated by studies that employ highly context-specific information from ground truth datasets (MacNeil et al. 2024), surveys (Tzachristas et al. 2025) and structured interviews (Zhang et al. 2025) as proxies for observed human variability and patterns. High alignment was achieved, albeit reliant on the inclusion of the expected results in the prompt (e.g., the taxonomy and frequency of found of issues) (MacNeil et al. 2024), or on hyper-tuned directives (e.g., guidance on how to interpret specific correlations between attributes) (Tzachristas et al. 2025). Even access to qualitative data from real individuals does not implicitly lead to psychometric alignment or realistic variability (Zhang et al. 2025; Wang et al. 2025d).

LLM may also assist directly with persona creation, particularly since their target users may be unable to define salient information (Benharrak et al. 2024). Often assessed for heuristic use by practitioners, generated persona may also serve as inputs for simulation. Despite subjective believability (Schuller et al. 2024; Salminen et al. 2024), even complex generated personas lack perceived realism, narrative richness and variability, with biases entangled in a variety of issues, both socio-ethical and fidelity-related (Venkit et al. 2025; Lazik et al. 2025; Barambones et al. 2024; Salminen et al. 2024). Manipulation of demographic representation does not mitigate these issues (Ma et al. 2024). Prompts that include context of use (e.g., design) can yield hyper-diverse and potentially hallucinatory personas (Ataei et al. 2025). Repetition in personas generated in parallel without mutual awareness can be mitigated by serial (single-prompt) generation (Ataei et al. 2025). Simulation of fully LLM-generated personas may serve in cases that prioritize utility over fidelity, such as in generation of pre-training data for machine learning when data is scarce (Leng et al. 2025).

To increase diversity and explainability and reduce biases, data-driven personas can be generated based on surveys and conversation transcripts (Jung et al. 2025; Wang et al. 2024). LLMs' lack of understanding and contextual awareness can be mitigated by involving humans in data grouping (Shin et al. 2024) and model creation (Wang et al. 2024), or by using clustering algorithms (Jung et al. 2025) (human-LLM collaboration vs. automation). Even using personas, LLMs can still struggle with representing complex human characteristics, abilities, needs and challenges (Sun et al. 2025b).

As is evident, there are limitations to simulation fidelity that LLMs can achieve with static personas, whether they employ high-level directives or model real observations. Cognitive modeling encompasses experimental approaches that go beyond the level of rudimentary prompt engineering by attempting to recreate the effects of human cognitive skills and processes instead of naive expectations for LLMs to manifest them latently (Owoicho et al. 2023; Algherairy and Ahmed 2025; Park et al. 2023; Wang et al. 2025b; Zhan et al. 2025; Gao et al. 2025; Ju et al. 2025; Amirizaniani et al. 2024). Some are task-specific, such as modeling information needs and patience in search systems (Owoicho et al. 2023). Others comprise comprehensive mental frameworks.

To reduce gaps between LLM outputs and multilayered human reasoning will require addressing flaws in causal, factual, deductive and abstract reasoning, as well factors that stem from lack of embodied subjectivity, including personal goals, intents, beliefs, abilities and emotions (Gao et al. 2025). Modeling emotions and intent yielded only limited improvements to the ability to reason about others' mental state (Theory of Mind) (Amirizaniani et al. 2024). While incorporating goals and intents structured in an ontology, simulated conversations are significantly lexically different from human utterances (Algherairy and Ahmed 2025).

In humans, memory is a set of processes that manage the persistent yet constantly shifting sum of experiences, thoughts and knowledge that shape each person's individuality. Approaches aimed at addressing LLMs' lack of personal memory (Park et al. 2023; Wang et al. 2025b; Zhan et al. 2025; Zhu et al. 2025) must contend with memory size, which can easily exceed the capacity of LLMs' context window (Park et al. 2023). Memory can thus be managed externally as a memory stream of observations, plans and reflections (Wang et al. 2025b), controlled by processes designed to mimic mechanisms known from cognitive neuroscience (Wang et al. 2025b). This includes retrieval of memories relevant in the current context based on their weight (Ju et al. 2025), but also periodic reflection to make abstract inferences, along with forgetting to suppress clutter (Park et al. 2023; Wang et al. 2025b). Limited-scope memory modeling achieved considerable strides in behavior simulation over baseline predictive models, although significant differences from real human behavior are still evident (Zhan et al. 2025).

Aside from personas and cognitive modeling of human subjects, modeling their environment (e.g., rooms at home, time of year and day) is a promising yet underexplored direction (Xu et al. 2025; Leng et al. 2025).

4.6 Model selection (RQ6)

RQ6: What LLMs are used to generate synthetic participants?

A variety of Large Language Models and parameter configurations were incorporated in the generation of synthetic participants, as demonstrated in Figure 6. A study on average utilized 3.28 models (SD=4.26). In the majority of studies, the parameters used were not clarified. The largest family of models, GPT, mostly consisted of GPT-4 (83), GPT-3.5 (78), GPT-3 (59), GPT-4o (35) and GPT-2 (4). In 27 cases, the GPT version was not provided. The most common LLaMA models were versions LLaMA-2 (44) and LLaMA-3 (46). The relevant articles are summarized in Table 7.

The behavior of models can differ significantly (Brand et al. 2024; Almeida et al. 2024; Gupta et al. 2025; Huang et al. 2025a; Kambhatla et al. 2025; Sethi et al. 2025; Gupta et al. 2025). However, due to prompt brittleness, stochastic unpredictability and contextual and temporal sensitivity, this heterogeneity tends to be highly context-specific. Combined with generally lackluster algorithmic fidelity (see 4.3 Issues), no general guidelines can be offered to recommend specific models as superior for specific simulation tasks. Since simulation diverges from LLMs' typical conversational objectives, caution should also be exercised when interpreting benchmarks (Gao et al. 2025). Good scores may not generalize outside of standard tasks, or be reflective of nuanced human behavior. Validation should be context-aware as any prompt or its change could trigger a model's failure mode.

The patterns are surprisingly inconsistent even while comparing models based on age and the number of parameters. Some studies assigned higher fidelity to the greater parameter count of newer models (e.g., GPT-4 better than GPT-3.5) (Petrov et al. 2024; Amirova et al. 2024; Xie et al. 2024; Brand et al. 2024). The difference can be negligible, favoring the low cost of smaller models (Fiori et al. 2024). This is inconclusive however, as smaller older models can also exhibit less distortion, including representation of human biases (Milička et al. 2024; Aher et al. 2023; Kwok et al. 2024; Gao et al. 2025). This contradicts the idea of newer, larger models becoming more human-like. Even the Centaur model – touted as being designed to replicate human cognition (Binz et al. 2025) – can fall short of its expectation (Gao et al. 2025). Ensemble methods may address the lack of within-model variability by combining the behaviors of different models, but it does not prevent algorithmic biases from manifesting (Liu et al. 2025).

Fine-tuning models on context-specific data can be valuable. Even small LLMs (e.g., Llama-8b, DistilGPT-2) can compete with or outperform larger models such as GPT-4 (He-Yueya et al. 2024;

Table 7 LLMs selected for participant simulation.

Model	Develop.	Publications
GPT	OpenAI	Abbasiantaeb et al. (2024); Abdel-Khalek et al. (2025); Aher et al. (2023); Al et al. (2024); Alaofi et al. (2023); Algherairy and Ahmed (2025); Almeida et al. (2024); Alonso et al. (2025); Amirizaniani et al. (2024); Amirova et al. (2024); Anderson et al. (2024); Argyle et al. (2023); Ataei et al. (2025); Bakaev et al. (2025); Barambones et al. (2024); Benedetto et al. (2024); Benharrak et al. (2024); Bhandari et al. (2024, 2025); Bickley et al. (2025); Binz and Schulz (2023); Bisbee et al. (2024); Bodroza et al. (2024); Boelaert et al. (2025); Braga et al. (2024); Brand et al. (2024); Busker et al. (2025); Cheng et al. (2023); Cho et al. (2024); Chuang et al. (2024); De et al. (2025); Dey et al. (2025); Dominguez-Olmedo et al. (2024); Dorner et al. (2023); Du et al. (2024); Elyoseph et al. (2023); Fang et al. (2024); Fiori et al. (2024); Flamino et al. (2025); Gao et al. (2025); Giorgi et al. (2024); Gmyrek et al. (2025); Gray et al. (2024); Griffin et al. (2023); Gu et al. (2025); Gupta et al. (2025); Hagendorff et al. (2023); Hämäläinen et al. (2022, 2023); Han et al. (2024); Hartmann et al. (2023); Havaei et al. (2024); He-Yueya et al. (2024); Herbold et al. (2023); Heyde et al. (2025); Heyman and Heyman (2024); Homer (2025); Hu et al. (2023); Huang et al. (2024a,b,c,d, 2025a); Huff and Ulakçı (2024); Hung et al. (2025); Hwang et al. (2023); Imschloss et al. (2025); Ivasciuc et al. (2025); Jalali and Akhavan (2024); Jiang et al. (2023, 2024); Jiaqi et al. (2025); Ju et al. (2025); Jung et al. (2025); Kalinin (2023); Kang et al. (2025); Kapania et al. (2025); Khalil et al. (2025); Khanshan et al. (2024); Kim and Lee (2024); Kitadai et al. (2024); Kovač et al. (2024); Kwok et al. (2024); Lazik et al. (2025); Lee et al. (2024a,b, 2025); Leng et al. (2025); Li et al. (2024a,b,c); Li and Qi (2025); Lim et al. (2025); Liu et al. (2025); Lo et al. (2025); Lu et al. (2025); Ma et al. (2024, 2025b); MacNeil et al. (2024); Manning et al. (2024); Marjieh et al. (2024); Matsumura et al. (2025); Milička et al. (2024); Molchanova et al. (2025); Mori et al. (2024); Motoki et al. (2024); Neumann et al. (2025); Niszczota et al. (2025); Nóbrega et al. (2025); Owoicho et al. (2023); Park et al. (2022, 2023); Park and Kim (2024); Park et al. (2024a,b); Petrov et al. (2024); Qi et al. (2025); Qu and Wang (2024); Ran et al. (2025); Robinson et al. (2023); Rozado (2023); Rutinowski et al. (2024); Salecha et al. (2024); Salminen et al. (2024); Sanders et al. (2023); Sannigrahi et al. (2024); Santurkar et al. (2023); Sattele and Ortiz (2024); Schuller et al. (2024); Serapio-García et al. (2025); Sethi et al. (2025); Shan and Jia (2024); Shin et al. (2024); Shrestha et al. (2024); Smrke et al. (2025); Sop and Kurcer (2024); Sparrenberg et al. (2024); Steinmacher et al. (2024); Suguri Motoki et al. (2024); Sun et al. (2024, 2025a,b); Tavast et al. (2022); Tjuatja et al. (2024); Trott (2024); Tzachristas et al. (2025); Venkit et al. (2025); Wang et al. (2024, 2025a,b,c,d,e); Winter et al. (2024); Xiang et al. (2024); Xie et al. (2024); Yang et al. (2025); Yoon et al. (2024); Zhan et al. (2025); Zhang et al. (2024a,b,c,d, 2025); Zhao et al. (2025); Zhou et al. (2025); Zhu et al. (2025)
LLaMA	Meta	Algherairy and Ahmed (2025); Almeida et al. (2024); Amirizaniani et al. (2024); Ball et al. (2025); Bernardelle et al. (2025); Bhandari et al. (2024, 2025); Bodroza et al. (2024); Boelaert et al. (2025); Brand et al. (2024); Cava and Tagarelli (2025); Cho et al. (2024); Chuang et al. (2024); Dey et al. (2025); Dominguez-Olmedo et al. (2024); Dorner et al. (2023); Du et al. (2024); Flamino et al. (2025); Gao et al. (2025); Gupta et al. (2025); He-Yueya et al. (2024); Huang et al. (2024a,b,c, 2025a,b); Jiang et al. (2023, 2024); Jiaqi et al. (2025); Ju et al. (2025); Kambhatla et al. (2025); Kang et al. (2025); Kim and Lee (2024); Kovač et al. (2024); Li et al. (2025); Lim et al. (2025); Liu et al. (2025); Ma et al. (2025a); Neumann et al. (2025); Qi et al. (2025); Ran et al. (2025); Salecha et al. (2024); Serapio-García et al. (2025); Sethi et al. (2025); Smrke et al. (2025); Suguri Motoki et al. (2024); Suh et al. (2025); Sun et al. (2025a); Tjuatja et al. (2024); Wang et al. (2024, 2025a,d); Xie et al. (2024); Yonekura et al. (2024); Zhang et al. (2024c); Zhao et al. (2025)
Claude	Anthropic	Almeida et al. (2024); Brand et al. (2024); Gao et al. (2025); Gupta et al. (2025); He-Yueya et al. (2024); Huang et al. (2025a); Hung et al. (2025); Kambhatla et al. (2025); Li et al. (2024c); Lu et al. (2025); Min and Jeong (2024); Molchanova et al. (2025); Salecha et al. (2024); Sethi et al. (2025); Zhang et al. (2024c,d, 2025)
Mistral	Mistral AI	Amirizaniani et al. (2024); Ball et al. (2025); Bernardelle et al. (2025); Cava and Tagarelli (2025); Huang et al. (2025a); Kambhatla et al. (2025); Kang et al. (2025); Kovač et al. (2024); Li et al. (2025); Marjieh et al. (2024); Neumann et al. (2025); Serapio-García et al. (2025); Suh et al. (2025); Sun et al. (2025a)
Gemini	Google DeepMind	Almeida et al. (2024); Bhandari et al. (2024); Bodroza et al. (2024); Cho et al. (2024); Du et al. (2024); He-Yueya et al. (2024); Huang et al. (2024a); Kang et al. (2025); Liu et al. (2025); Sethi et al. (2025); Smrke et al. (2025); Venkit et al. (2025); Zhang et al. (2024d, 2025)
Qwen	Alibaba Cloud	Ball et al. (2025); Bernardelle et al. (2025); Dey et al. (2025); Kambhatla et al. (2025); Kang et al. (2025); Kovač et al. (2024); Li et al. (2025); Qi et al. (2025); Sun et al. (2025a); Zhang et al. (2024c,d); Zhao et al. (2025)
Mixtral	Mistral AI	Bodroza et al. (2024); Boelaert et al. (2025); Cava and Tagarelli (2025); Huang et al. (2024b); Kang et al. (2025); Kovač et al. (2024); Ma et al. (2025a); Molchanova et al. (2025); Serapio-García et al. (2025)
DeepSeek	DeepSeek	Huang et al. (2025a); Jiaqi et al. (2025); Sethi et al. (2025); Venkit et al. (2025); Zhang et al. (2024d)
T5	Google AI	Anderson et al. (2024); Chuang et al. (2024); Jiang et al. (2023); Serapio-García et al. (2025); Sun et al. (2025a)
Gemma	Google DeepMind	Ball et al. (2025); Cava and Tagarelli (2025); Ma et al. (2025a); Zhao et al. (2025)

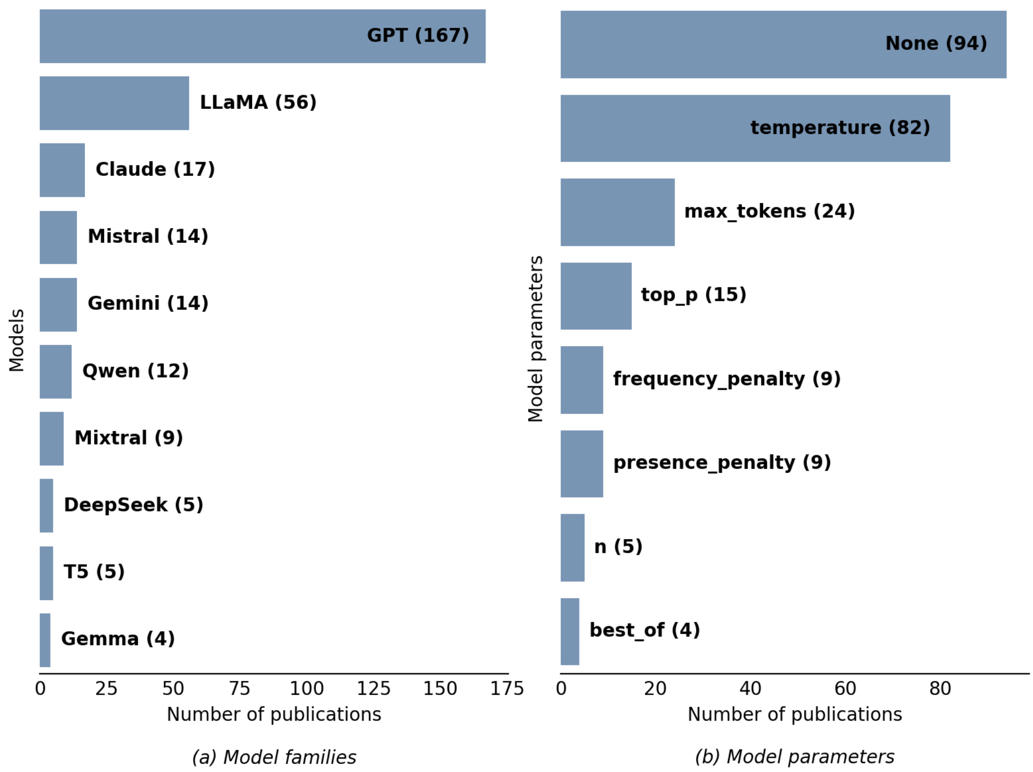

Fig. 6 Use of LLMs in participant simulation studies grouped by (a) model family and (b) parameters. Not displayed here are models and parameters represented by three articles and less.

Khalil et al. 2025) when custom-trained on relevant samples. This may still lead to only reductively mimicking the training data however (Gao et al. 2025). Furthermore, fine-tuning with real results taken from an identical task but a different context (e.g., product) can have a negative impact on generalization (Brand et al. 2024).

LLM parameters like temperature, top-p, presence penalty or frequency penalty have minimal observable effects on simulation (Barambones et al. 2024; Li and Qi 2025). Increasing the randomization of tokens through temperature can have no meaningful impact on uniformity (Petrov et al. 2024) and thematic diversity (Ma et al. 2025b). Deterministic temperature of 0 was suggested by (Li and Qi 2025) to improve replicability of research, although doing so may not authentically represent the properties of stochastic models. Multiple runs at higher temperatures may be preferred for robust evaluations (Ma et al. 2024; Jiaqi et al. 2025; Algherairy and Ahmed 2025).

5 Discussion

Our literature review spotlights a spark of optimism toward synthetic participants that pervades a variety of fields and scenarios (Zhao et al. 2025; Zhu et al. 2025; Li et al. 2024b; Xie et al. 2024). While such sentiment may be justified, a cautious line should be drawn about framing simulations as valid analogues or substitutes for humans without clear restrictions (Argyle et al. 2023; Li et al. 2024b; Xie et al. 2024; Smrke et al. 2025; Benedetto et al. 2024; Wang et al. 2025b; Zhao et al. 2025; Zhu et al. 2025; Park et al. 2022; Braga et al. 2024). Not only are such claims empirically precarious and actively warned against by authors (Venkit et al. 2025; Gao et al. 2025; Trott 2024; Amirova et al. 2024; Wang et al. 2025a; Petrov et al. 2024; Ma et al. 2024; Mori et al. 2024; Almeida et al. 2024; Boelaert et al. 2025; Lu et al. 2025; Park et al. 2024b; Zhang et al. 2024b; Park et al. 2022; Brand et al. 2024; Park et al. 2024a; Smrke et al. 2025), they echo socially harmful replacement narratives linked to broader ethical concerns, such as withholding credit from obscured and undervalued human labor that AI models depend on for training and fine-tuning (Crockett and Messeri 2023). For constructive realization of the transformative potential of synthetic participants (Ju et al. 2025; Alonso et al. 2025; Wang et al. 2025b), relationships and similarities to humans need to be considered with contextual nuance.

Simulations exhibit a direct relationship between algorithmic fidelity and costs (Dahlstrom et al. 2009), and synthetic participants are not an exception. Their costs accumulate beyond the preexisting excessive operational costs of LLMs (Chen et al. 2025). Approaches with better human alignment rely on context-specific empirical data (incorporated in pre-training, RAG or few-shot prompting) or steering with personas and cognitive modeling (He-Yueya et al. 2024; Tzachristas et al. 2025; Zhan et al. 2025; Gao et al. 2025). Therefore, practitioners need to weigh utility based on the fidelity-cost tradeoff. More sophisticated simulations require greater resources (e.g., to collect relevant data as the ground truth to construct and validate personas). Methods will need to focus on reusability, dynamicity and updatability to improve their cost value.

Although better prompting and fine-tuning are essential for improving simulations (Barambones et al. 2024; Sun et al. 2025b), they should be viewed as patches for specific goals rather than solutions that can be combined to obtain agents that are holistically human-like. Parameter manipulation does not address fundamental issues with models. It cannot eliminate simulation-induced biases, only redistribute them. The generalizability of new unvalidated findings to humans, which depends on the simulation's algorithmic fidelity (Amirova et al. 2024), becomes anchored to biases that the manipulation may not address due to real world complexity. As a parallel to participant priming (which may invalidate some research by artificially distorting its conclusions (Krajcovic et al. 2025)), further biases may even be introduced. Multifaceted manipulations (e.g., personas modeling demographics, lifestyle, habit and health (Leng et al. 2025)) should be assessed through ablation to isolate the effects of individual factors.

The surveyed literature exposes a recurring struggle of research to select measures that are adequate for rich evaluation of algorithmic fidelity. By using the same validity measures traditionally used for human data, synthetic responses can sometimes be interpreted out of context as "surpassing" human responses, since they can appear more consistent (Wang et al. 2025e), informative (Hung et al. 2025), complex and lexically diverse (Herbold et al. 2023; Sethi et al. 2025), along with containing more topics (Zhang et al. 2024b). However, such measures offer limited perspectives, their values potentially inflated because they collate elaborative depth – linguistically rich padding of homogeneous stereotypes and stochastic parroting, hallucinated potentially out of context – with real latent depth that is subjective, as well as contextually and narratively aware (Venkit et al. 2025; Lazik et al. 2025; MacNeil et al. 2024; Bakaev et al. 2025; Sanders et al. 2023).

Therefore, aside from alignment, fidelity validations should incorporate richer measurement frameworks. These should be able to signal divergences from human observations and their experiential subjectivity, as well as support their explanations. Relevant measures currently include creativity features such as surprisal and semantic diversity, as well as other variability-based constructs (Venkit et al. 2025; Bakaev et al. 2025; He-Yueya et al. 2024). Although Venkit et al. (2025) originally discussed this as a priority with respect to minority representation in generated personas, these concerns extend more generally – e.g., to quantitative responses with generated reasoning and latent factors – given the prevalence of behavioral and cognitive distortions (see 4.3 Issues).

To illustrate narrative depth and semantic variability of text as key evaluation measures, we present an example involving simulation of personality. LLMs encode psychological theory. They can complete personality assessments and generate sentences that align with the standard understanding of personality traits (Jiang et al. 2024; Molchanova et al. 2025; Han et al. 2024). However, models are inherently simplified. Personality is complex, self-contradictory, situational and constantly affected by day-to-day experiences (Toomela 2025). Simulations that aim to approximate the behavioral implications of personality will demand validation of their ability to model an internally consistent, subjective, and experiential narrative.

Given their fundamental differences from humans, it is more valid to construe synthetic participants as a heuristic approach, with heuristics implicitly and intransparently represented via the models' embeddings and mechanisms (Li et al. 2024b; Brand et al. 2024; Xiang et al. 2024). This heuristic-like generic nature is mechanistically inherent to processes by which LLMs average information and assimilate minority perspectives (Jung et al. 2025). Further affirming this is the fact that even Large Reasoning Models (LRMs, newer iterations of LLMs designed for reasoning through self-reflection) appear to operate heuristically in reasoning problems — either they find a correct solution to a simple problem early and then they just overthink it, or the problem is complex and their reasoning collapses (Shojaee et al. 2025). In terms of variability, synthetic participants have more in common with personas (Gu et al. 2025), i.e. fictional characters used as storytelling devices (Lazik et al. 2025; Vranić et al. 2024) rather than real people. Their resemblance to human-crafted personas can vary (Schuller et al. 2024; Shin et al. 2024; Gu et al. 2025; Lazik et al. 2025), but both provide practically-oriented abstractions balanced by a propensity for biases and stereotypes – expert-imposed or algorithmic (Schuller et al. 2024).

As heuristic tools, the coverage of LLM-driven solutions can be broad, such as extending to edge cases that human participants may not think about (Xiang et al. 2024). As a caveat, traditional personas tend to be intentionally shallow, supporting deliberative empathy, whereas the high believability of synthetic participants gives them the trappings of empirical data, promoting expectations for detailed human-like interactions and a personhood bias (Gu et al. 2025). Amplification of practitioners' own flawed assumptions (Lu et al. 2025; Dahiya and Kumar 2021) can contribute to de-skilling, especially by non-technical professionals unaware of technical limitations. The heuristic framing can be used to set more appropriate expectations about LLMs' restrictions (Gu et al. 2025; Kaate et al. 2025).

Adopting the heuristic lens, synthetic participants may be helpful for actionable insights in scenarios where empirical data is not available, scarce, protected by privacy, or would be disproportionately labor-intensive and costly to obtain within reasonable logistical constraints (Park et al. 2022; Yang et al. 2025; Benharrak et al. 2024; Gu et al. 2025). Beneficial scenarios include explorative studies that precede established knowledge of the target audience (Benharrak et al. 2024), interrogations of hypotheses (Brand et al. 2024; Alonso et al. 2025), investigations of the plausible effects of variables on complex systems (Park et al. 2022; Lim et al. 2025), low-stakes assessments (Liu et al. 2025) and piloting tasks and instructions before involving humans (Heyman and Heyman 2024; Brand et al. 2024). Over-reliance can lead to mistaken assumptions, confirming the relegation to a supplemental role (Dillion et al. 2023). For studies interested in unexpected incongruities, non-generic minority ideas, or details about exceptional people and exceptional scenarios (Ataei et al. 2025), algorithmic heuristics are too constraining as they cannot capture latent cognitive, behavioral, social and environmental complexity (Liu et al. 2025; Zhang et al. 2024b; Petrov et al. 2024; Lu et al. 2025; Jung et al. 2025).

The high performance of an augmentative solution that not only incorporates real participant data, but also its human cognitive preprocessing supports the case for human-AI collaboration over full machine autonomy (Shin et al. 2024). The benefit is bilateral as besides improving model performance, it also enhances practitioners' understanding of data (Shin et al. 2024). Methodological forays into keeping humans in the loop are few and still in their early stages – e.g., suggested interactive correction of hallucinations by Wang et al. (2025b) or control over the simulation through plugins by Zhu et al. (2025). However, their prospect for future study is significant for solving several key issues.

Practitioners may adaptively improve the heuristic value of synthetic participants by incorporating even small amounts of existing human data (Brand et al. 2024; Wang et al. 2025c; Trott 2024; Kitadai et al. 2024; Park et al. 2024a; Bhandari et al. 2024). For quantitative analysis, extrapolation from existing data along with transfer learning may reduce the effect of machine biases (Brand et al. 2024; Wang et al. 2025c), as well as expand available datasets and statistically enrich small ones ($N<10$) through hybridization (Bhandari et al. 2024; Trott 2024; Leng et al. 2025). For qualitative analysis, hybrid approaches may suggest plausible (not necessarily accurate) answers that were not provided during the original data collection, or to reduce the burden on participants (Kitadai et al. 2024; Park et al. 2024a)

In applied research, complementing or outperforming a baseline can present a practical advantage despite not authentically mirroring humans or imparting novel understanding (Bhandari et al. 2024). For machine learning models that are already dependent on synthetic data to mitigate data scarcity, LLM-generated data may be useful in contexts like automated speech recognition (Sannigrahi et al. 2024), question answering (Braga et al. 2024), or real-time prediction of behaviors in physical and digital settings (Zhan et al. 2025; Leng et al. 2025). Technologies like generative adversarial networks (GAN) can be more computationally efficient, but LLMs are promising because of higher flexibility (Khalil et al. 2025). As conversational agents, they may complement professional training of sensitive social interactions (e.g., between the doctor and the patient) (Wang et al. 2024), or reduce the cost of evaluating other conversational systems (Owoicho et al. 2023; Li et al. 2024c).

Training models on synthetic participants is best framed as a solution to the cold start problem – it can improve prediction early on, but have diminishing returns compared to human data (Leng et al. 2025). Therefore, pre-training can make effective use of datasets broadened with synthetic data, but human data is needed for fine-tuning to reflect real variability (Khanshan et al. 2024). Data can be fully synthetic, generated to structurally and statistically maintain the properties of sparse real samples without altering existing records (Khalil et al. 2025). By assigning synthetic features to real participants (e.g., behaviors, responses), data can also be enriched, rendering them semi-synthetic (Xu et al. 2025; Braga et al. 2024; Owoicho et al. 2023).

In spite of synthetic participants offering flattened representations of reality (Wang et al. 2025a), they can be conceptualized as useful in the niches of automated heuristics and applied research. Expansion of methods for enriching synthetic participants through personas together with modeling of cognitive processes, behavior and the environment (see 4.5 Participant modeling) are promising areas for future research to mitigate some of the issues discussed in this literature review (see 4.3 Issues). Nonetheless, with the growing concerns about LLMs, including the buildup of cognitive debt in their users due to over-reliance (Kosmyna et al. 2025), as well as negative social, economical and environmental impacts, from shortages and infrastructural risks caused by high energy and hardware consumption[8][9], to reinforcing discrimination (Kirk et al. 2024; Luitse and Denkena 2021), research should focus on designing human-in-the loop methodologies and interrogate efficient alternatives that can offer comparable value.

6 Threats to validity

The rigorous methodology of our systematic literature review aimed to systematically address most of the cardinal threats to our findings' validity: internal, external, construct and conclusion (Zhou et al. 2016). This includes mitigation of search biases though our hybrid database and reference search with iterative tuning of keywords. The primary focus of our search was on participant simulation. As such, our corpus may not contain some related works investigating the capabilities of LLMs in a general sense, since this would be out of scope for our review (e.g., Theory of Mind evaluation frameworks cited by Amirizaniani et al. (2024)).

To reduce the effects of publication bias, we dedicated balanced attention to reporting of issues and qualitative findings as to statistically significant and positive results, which tend to receive greater prominence. Coding, analysis and interpretation of findings was conducted by multiple researchers to reduce individual biases. The generalizability of our findings originates from a coverage of high-quality academic publications and preprints from a broad range of fields – including cognitive and behavioral psychology as lenses informing our reading of the current literature. Future advancements of language models and new methodologies aimed at simulating humans may extend the presented findings.

7 Conclusion

In this comprehensive multi-disciplinary literature review, we analyze the potential of state-of-the-art LLM-driven methods to generate synthetic participant data. We present an in–depth exploration of factors that differentiate LLMs from the cognitive processes of human participants. Then, we systematically categorize the types of issues encountered with synthetic participants and interrogate the ability of various approaches to address them. Our contributions culminate in a conceptual deconstruction of synthetic participants as surrogates for human participants. We offer the following implications for scholars and practitioners:

- LLM-generated synthetic participants are inadequate as representations of human cognition and behavior. They are mired by a variety of distortions (e.g., shallowness, genericness, invariability, stereotypicality, inconsistency, hyper-accuracy, hallucinations).
- Distortions are caused by inextricable differences from humans on mechanistic level (stochastic prediction of memorized online data fine-tuned to user expectations vs. subjective, experiential intelligence driven by biology and senses). Cognitive differences and lack of latent traits (e.g., personality as a product of nature and nurture, connected to memory) lead to significant downstream effects.

[8] IDC's report on the global memory shortage started in 2025: https://www.idc.com/resource-center/blog/global-memory-shortage-crisis-market-analysis-and-the-potential-impact-on-the-smartphone-and-pc-markets-in-2026/
[9] International Energy Agency's report on Energy & AI: https://www.iea.org/reports/energy-and-ai

- Prompts incorporating personas and cognitive modeling may improve algorithmic fidelity incidentally and to a limited degree, but depend on priming because of unreliable application of subtext and a lack of deep subjectivity.
- Evaluations synthetic participants should assess creativity, structure, semantic variability and depth of reasonings/justifications. The distinguishing value of lexical and believability metrics is limited.
- As a supplement to real data, synthetic participants are comparable to automated heuristics. Future approaches should explore augmentation to mitigate high believability and personhood bias.
- For data-driven applications, synthetic participants may address the cold start problem, but their contributive value can plateau rapidly.
- Pre-training or few-shot prompting with human data may improve algorithmic fidelity and variability.
- Models cannot be selected as preferable without validation. Large new models are not inherently more human-aligned, and can be inferior to smaller models pretrained on context-specific data.

Through an apt management of expectations, LLM-generated participant data can be useful in some applied niches. However, LLMs are ultimately just text generators. Based on their training, they predict which words are the most likely to come in sequence. This places them in stark contrast with real participants, who are involved in research for a simple reason: the layers of hidden complexity that can make them say or do seemingly unlikely things. The quintessence of research should always lie in the discovery of new facts, interpretations and insights. Homogenized generated text may seem tempting to trust, much like people are attracted to averages of human faces with more symmetrical anatomical features (Amaya et al. 2022). However, both offer only a distorting mirror of humanity and its idiosyncrasies.

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Declarations

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Conflict of interest. The authors have no relevant financial or non-financial interests to disclose.

Data availability. The publication dataset as well as extended results are openly available in a public paper repository at https://github.com/synthetic-users-research/synthetic-participants.

Author contribution. *Eduard Kuric:* Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing; *Peter Demcak*: Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing; *Matus Krajcovic*: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Appendix A Prompts

Eligibility screening prompt:

```
   Analyze the titles, abstracts, and keywords in the provided CSV. Identify
publications that involve synthetic participants in user or behavioral research,
where simulated responses or behaviors are used in place of real human subjects.
It concerns studies that gather simulated insights (e.g., LLM-generated survey
responses or synthetic respondents in usability testing testing) without involving
actual participants (without actual real people) or studies that compare this
synthetic data with human data. Return only the IDs of relevant publications,
separated by commas. No additional text.
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Appendix B Preprint quality checklist

The checklist contains 19 dichotomous items. The passing score was 15/19 or higher. This threshold was determined through qualitative exploration and comparison with published articles and papers also included in the literature review.

- Does the abstract clearly summarize the research focus, study design, methods, key findings and conclusions?
- Are the research objectives clearly stated and research questions clearly defined (e.g., RQ1: question)?
- Are the research hypotheses explicitly and clearly defined (e.g., H1: hypothesis)?
- Is the experiment process clearly described (e.g., steps taken, participant flow, experiment settings)?
- Are the outcome measures or variables clearly defined?
- Are the specific LLM models (e.g., GPT-4) and their parameters (e.g., temperature) clearly defined?
- Are real human participants involved in evaluating the results and comparing them with those generated by LLMs?
- Is the participant sample size reported?
- Is there a scientific method for determining the sample size (e.g. G*Power use)?
- Are sample demographics reported in detail (e.g., age, gender)?
- Are the statistical methods explicitly defined (e.g. statistical tests, groups being compared)?
- Are p-values and effect sizes reported for the statistical tests?
- Are descriptive statistics (e.g., means, standard deviations) reported?
- Are the main findings or implications presented and interpreted?
- Are the study's limitations discussed (e.g., biases, threats to validity, generalizability or uncertainties), preferably in a dedicated section?
- Are the study's future directions discussed (e.g., future work), preferably in a dedicated section?
- Do the authors disclose funding sources (e.g. a funding statement or acknowledgements)?
- Do the authors include a conflict of interest statement?
- Do the authors include a statement describing if and where study data are available? (e.g., URL link, supplementary materials)?

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